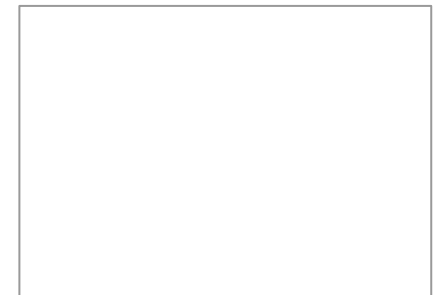




Quantum Database Search: Grover's Algorithm

Finding a needle in a haystack

Amlan Mukherjee



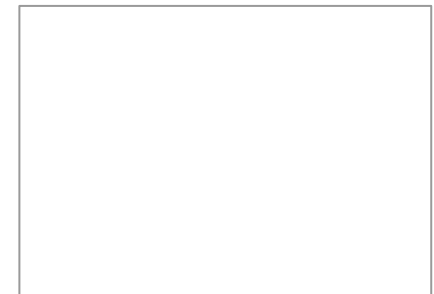
Amlan Mukherjee

Quantum Database Search: Grover's Algorithm

Finding a needle in a haystack



Lov K. Grover





Quantum Database Search: Grover's Algorithm

- Introduction : unstructured search
- Digital haystack
- Concepts: two steps
- Two steps with qubits
- Implementation
- Applications
- Conclusions



Needle in a haystack
(Unstructured Data)

Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

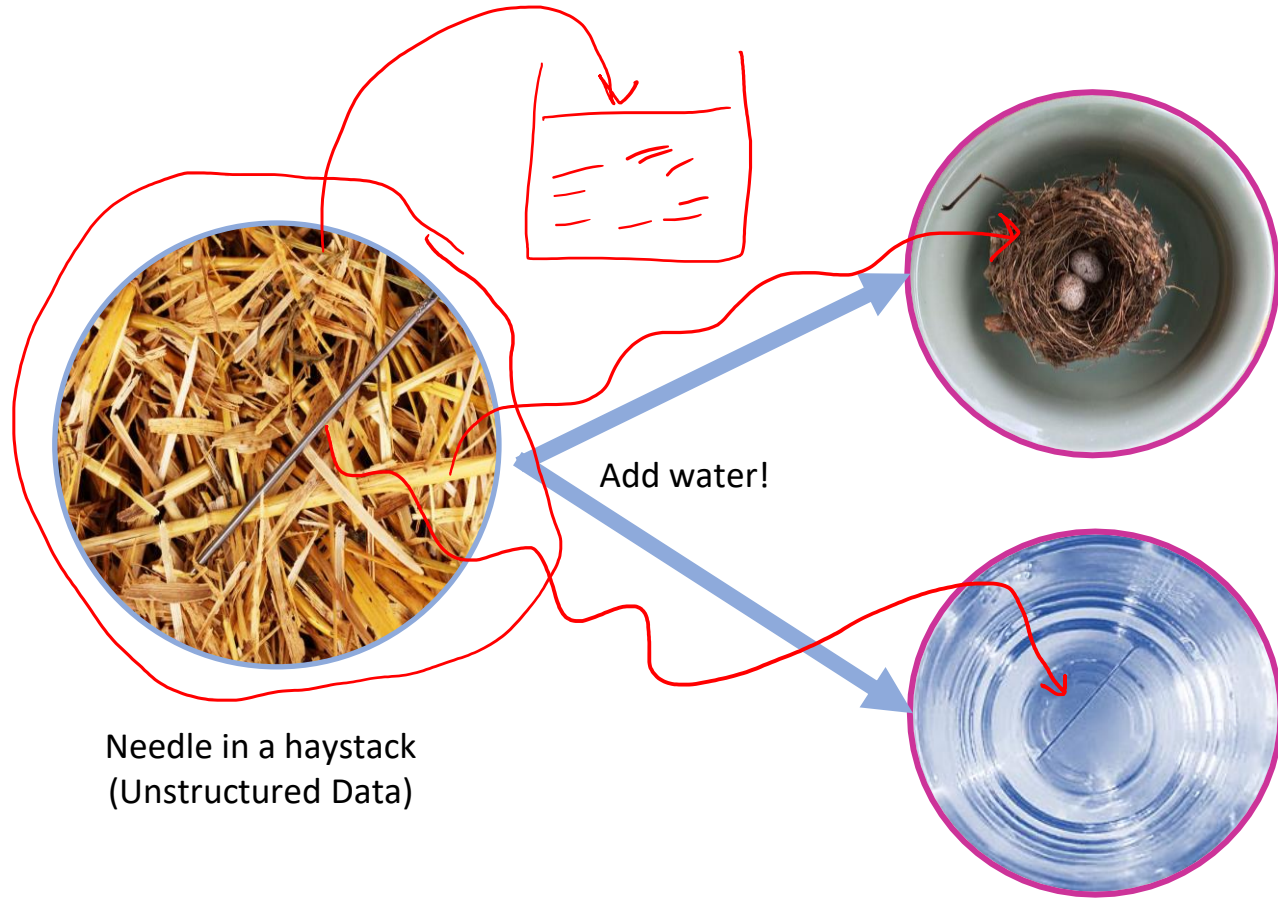


Needle in a haystack
(Unstructured Data)



Classical search can be inefficient!

Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)



Needle in a haystack
(Unstructured Data)

Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)



Needle in a haystack
(Unstructured Data)

Add water!



Magnet



Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

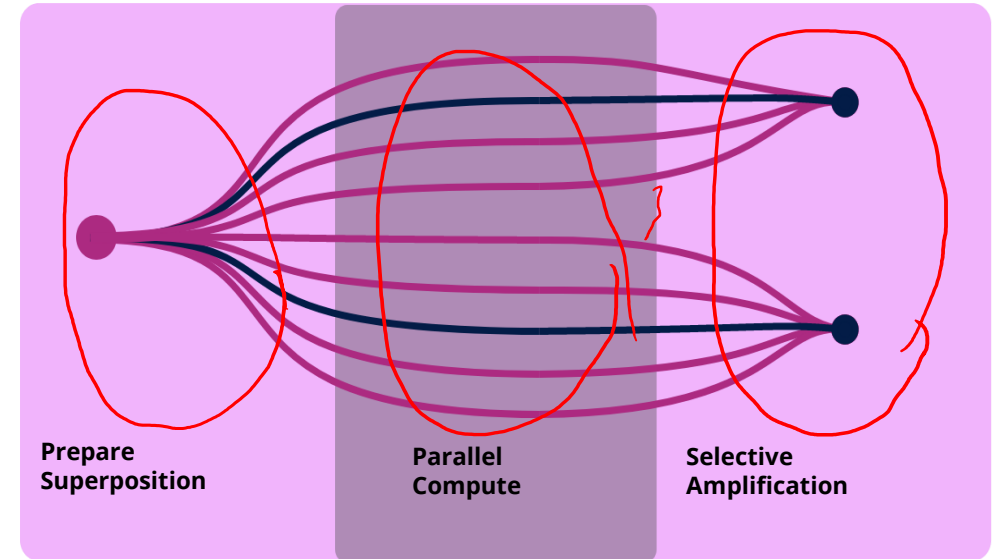


Needle in a haystack
(Unstructured Data)

Add water!



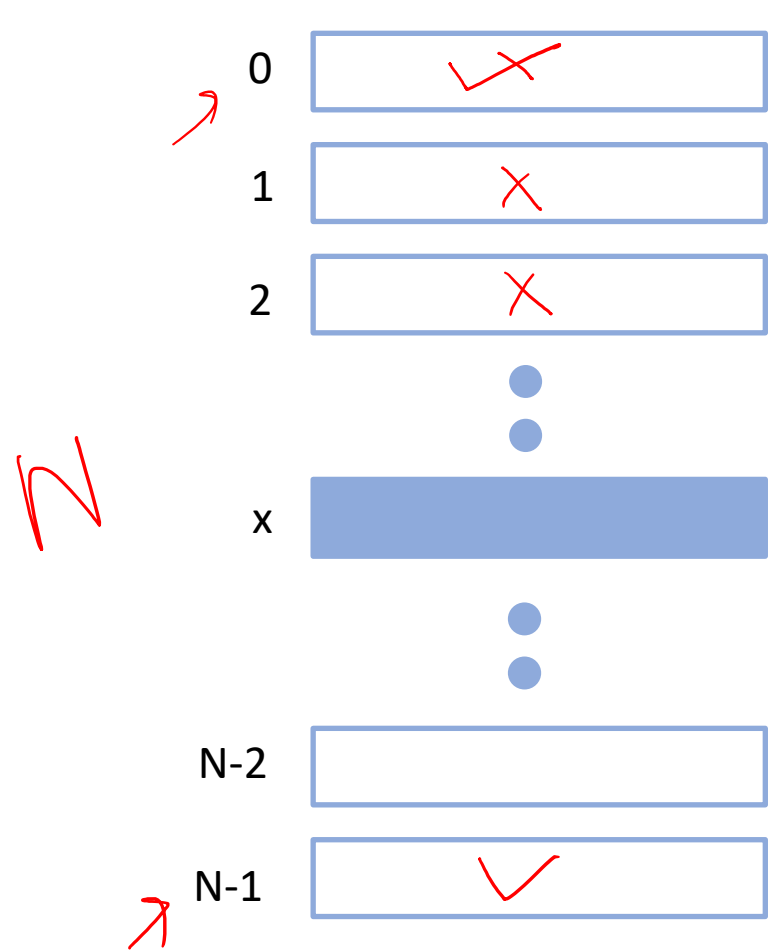
Magnet



Quantum ?

Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

Unstructured search



“Digital haystack”

Problem Statement:

$$f: \{0, 1, 2, \dots, N-2, N-1\} \rightarrow \{0, 1\}$$

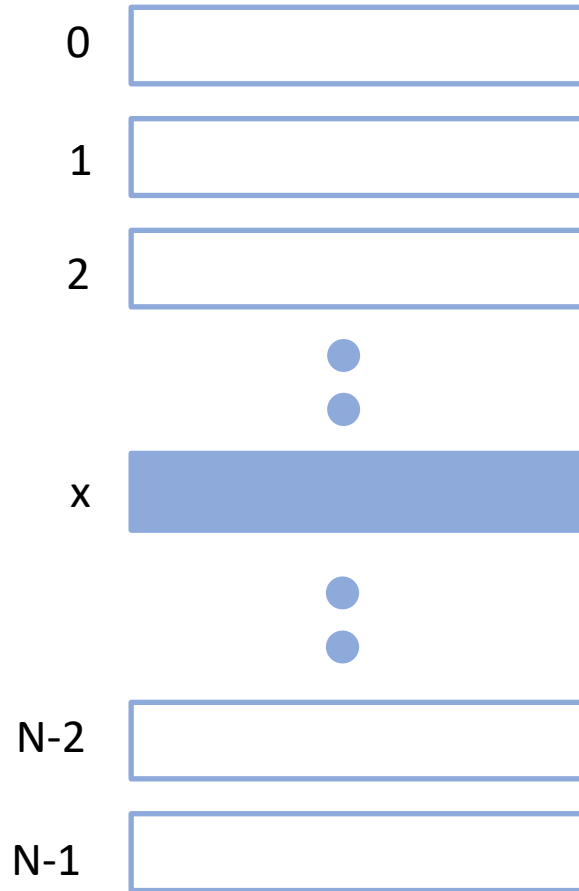
Find $x: f(x) = 1$

The hardest case: only one such $x: f(x) = 1$
(a needle!)

$$\Theta(N/2)$$

$$\Theta(\sqrt{N})$$

Unstructured search



“Digital haystack”

Problem Statement:

$$f: \{0, 1, 2, \dots, N-2, N-1\} \rightarrow \{0, 1\}$$

Find $x: f(x) = 1$

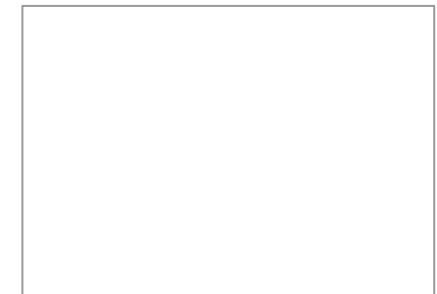
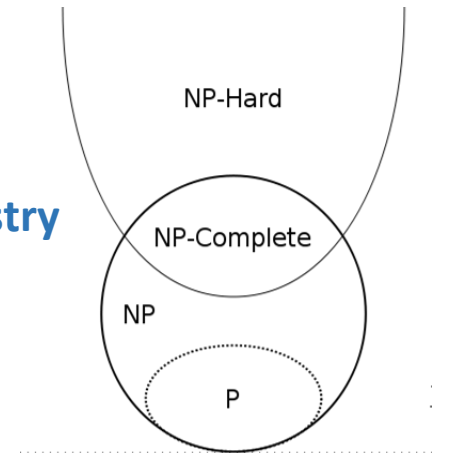
NP- complete class of problems:

ubiquitous in science, technology, and industry

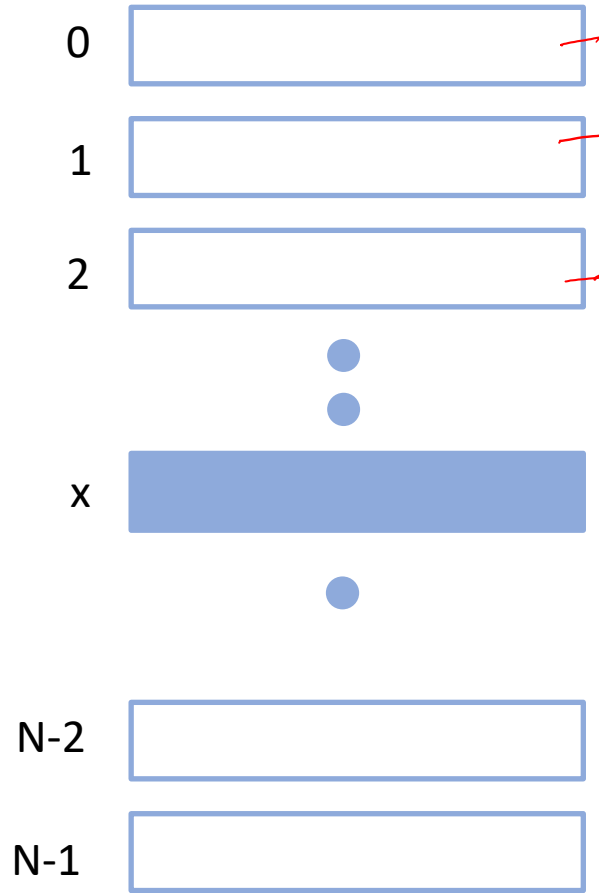
Satisfiability problem: Given a boolean formula in conjunctive normal form, for example:

$$(x_1 \vee \overline{x_2}) \wedge (x_1 \vee x_3 \vee \overline{x_4}) \wedge (\overline{x_2} \vee \overline{x_3}) \wedge (\overline{x_4})$$

Find an assignment that satisfies it (=1) if such an assignment exists.



Unstructured search



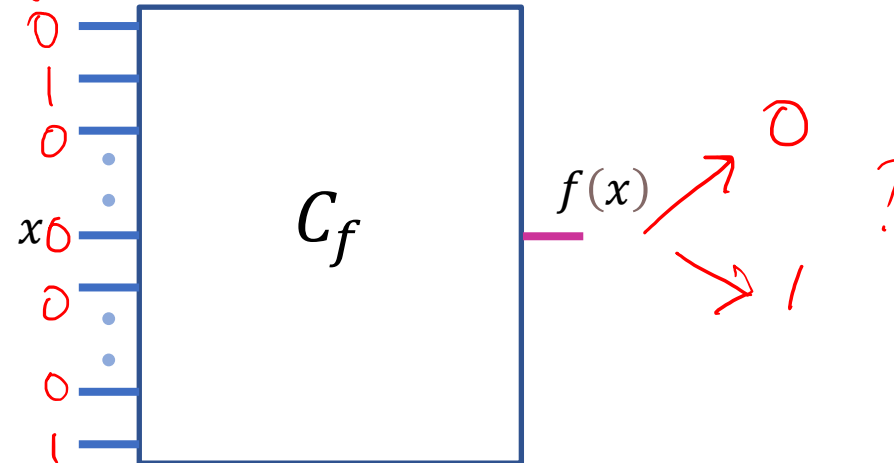
“Digital haystack”

Problem Statement:

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Find $x: f(x) = 1$

The hardest case: only one such $x: f(x) = 1$
(a needle!)

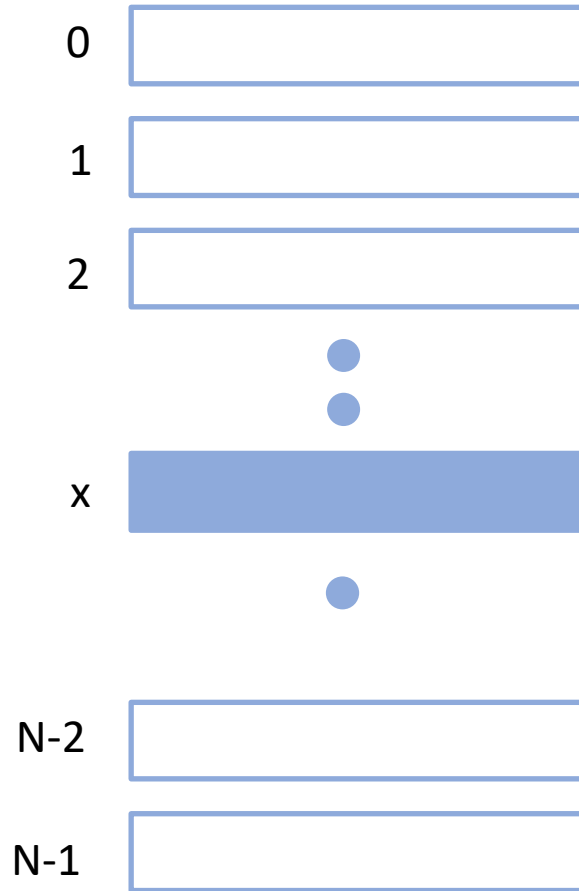


$N/2$ Steps

There are $N = 2^n$ configurations

Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

Unstructured search



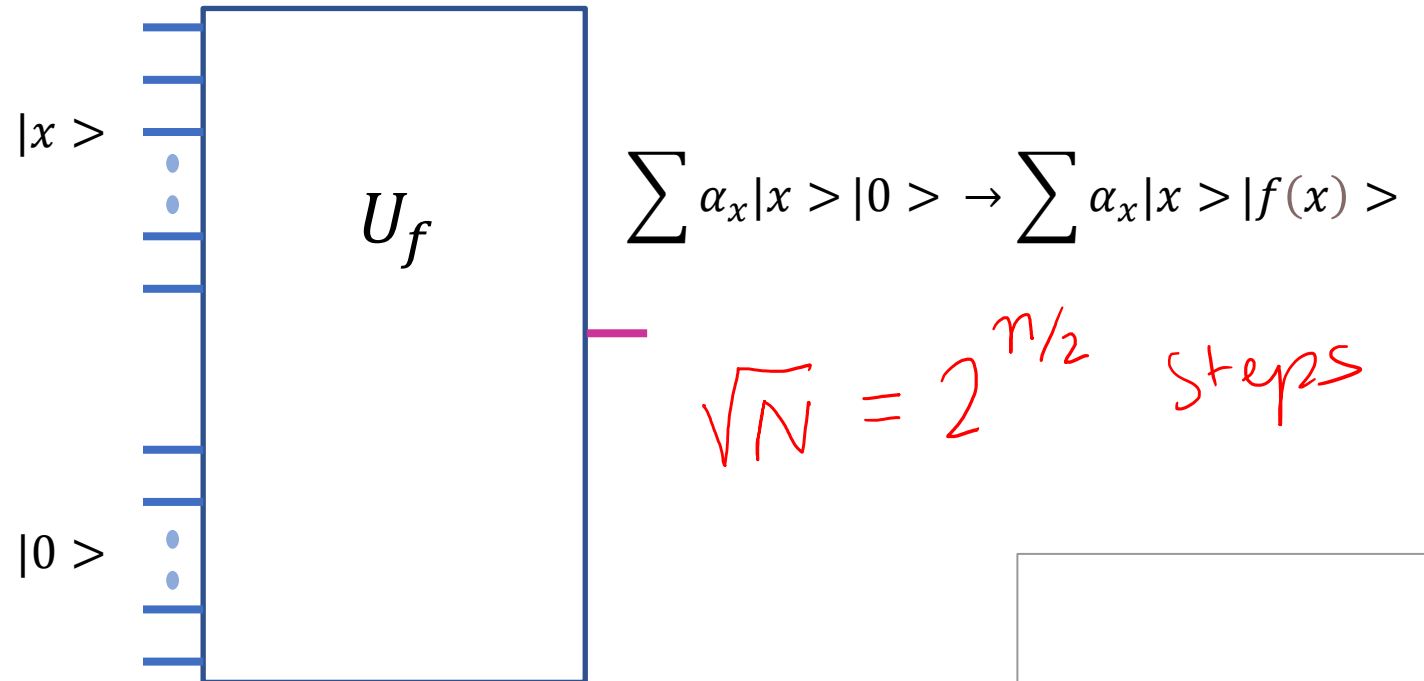
“Digital haystack”

Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

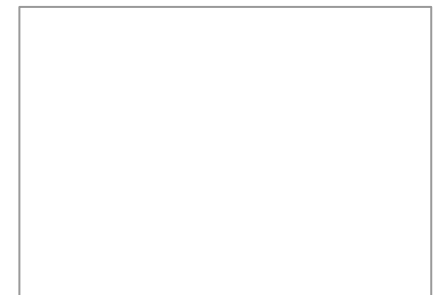
Problem Statement:

$$f: \{0, 1, 2, \dots, N-2, N-1\} \rightarrow \{0, 1\}$$

Find $x: f(x) = 1$ (The hardest case: only one such $x: f(x) = 1$)

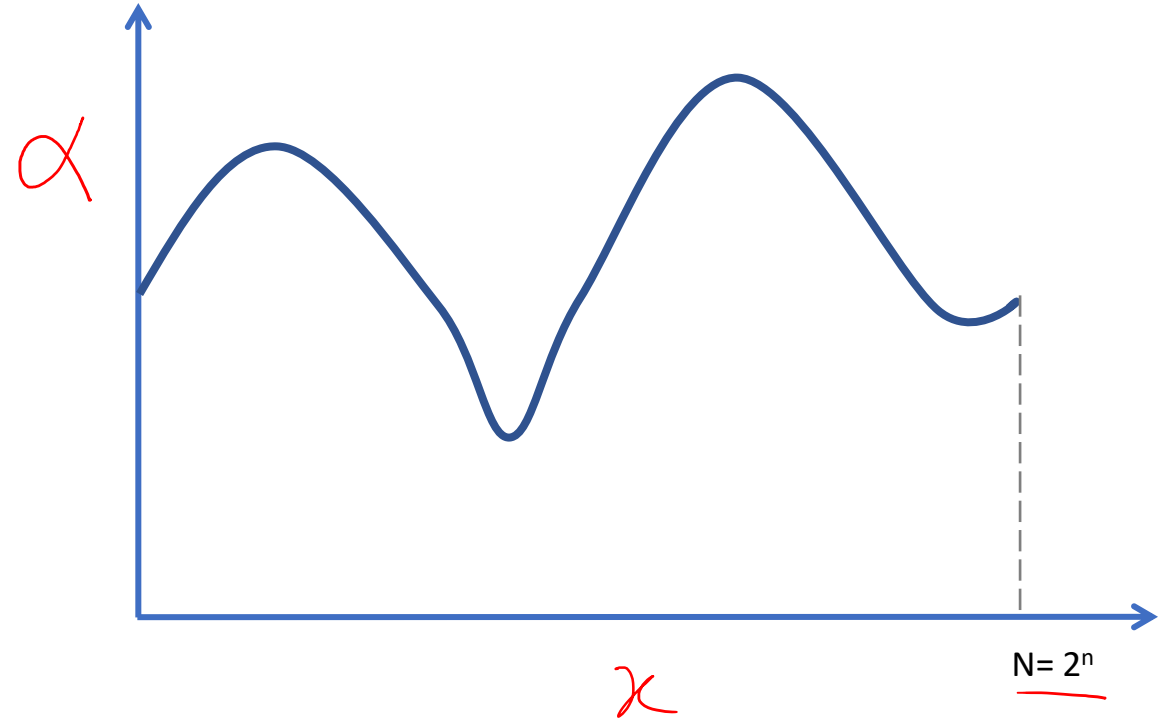


There are $N = 2^n$ configurations



Two steps: 1. Phase inversion

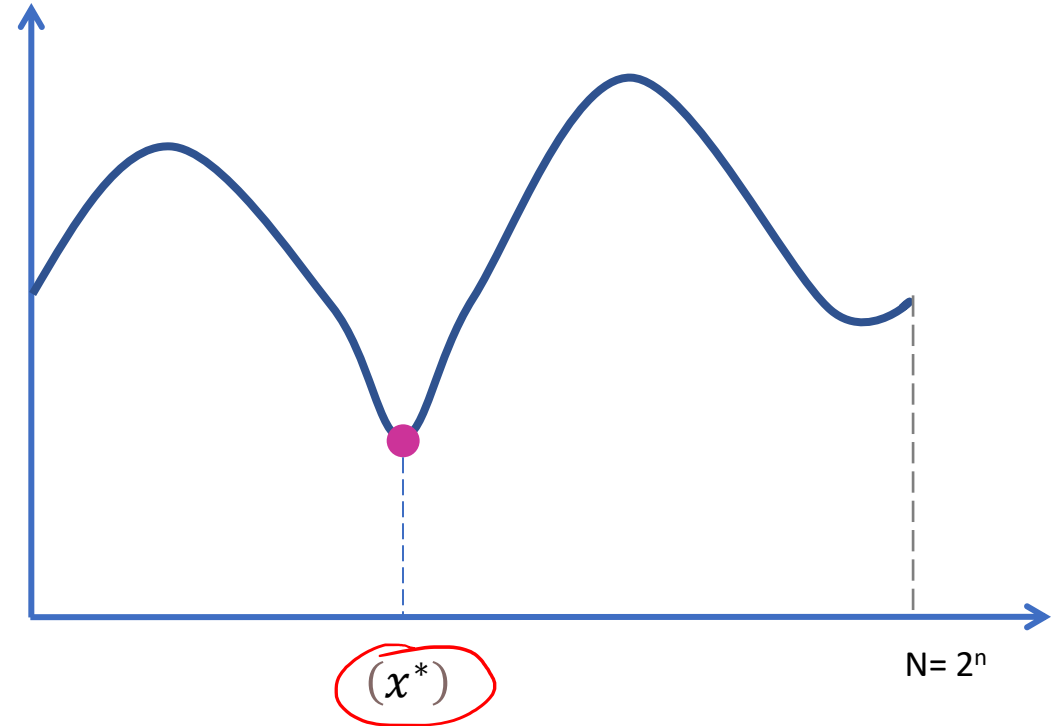
$$\sum_x \alpha_x |x\rangle$$



Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

Two steps: 1. Phase inversion

$$\sum_x \alpha_x |x\rangle$$



$$\underline{f(x^*) = 1}$$

Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

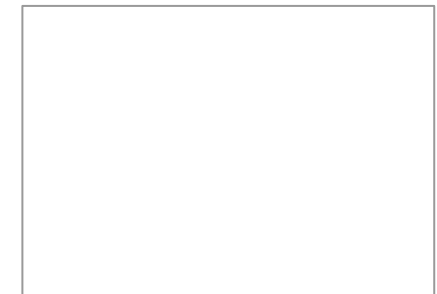
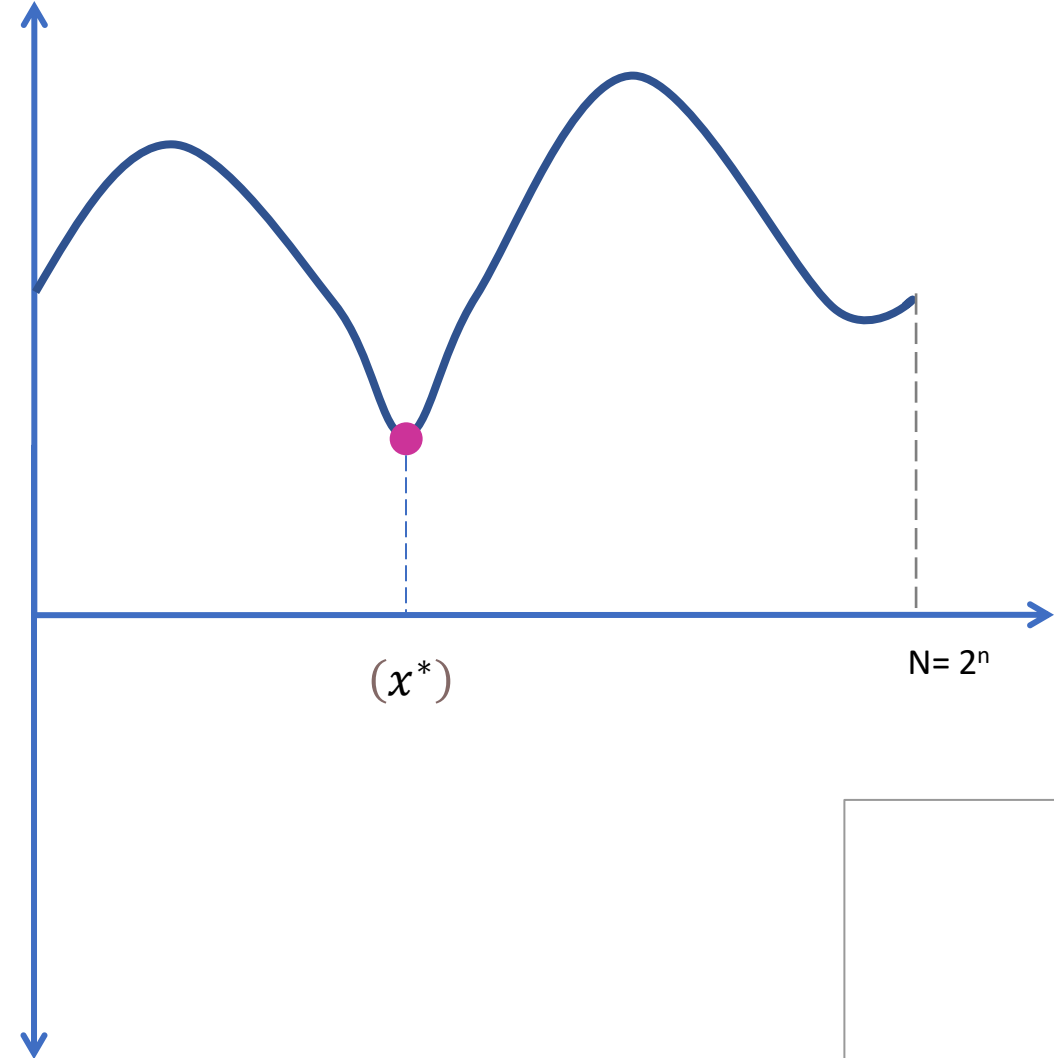
Two steps: 1. Phase inversion

$$\sum_x \alpha_x |x\rangle$$



$$\sum_{x \neq x^*} \alpha_x |x\rangle - \alpha_{x^*} |x^*\rangle$$

Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)



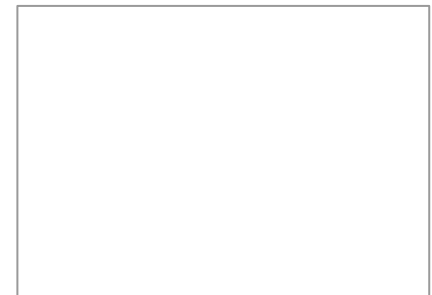
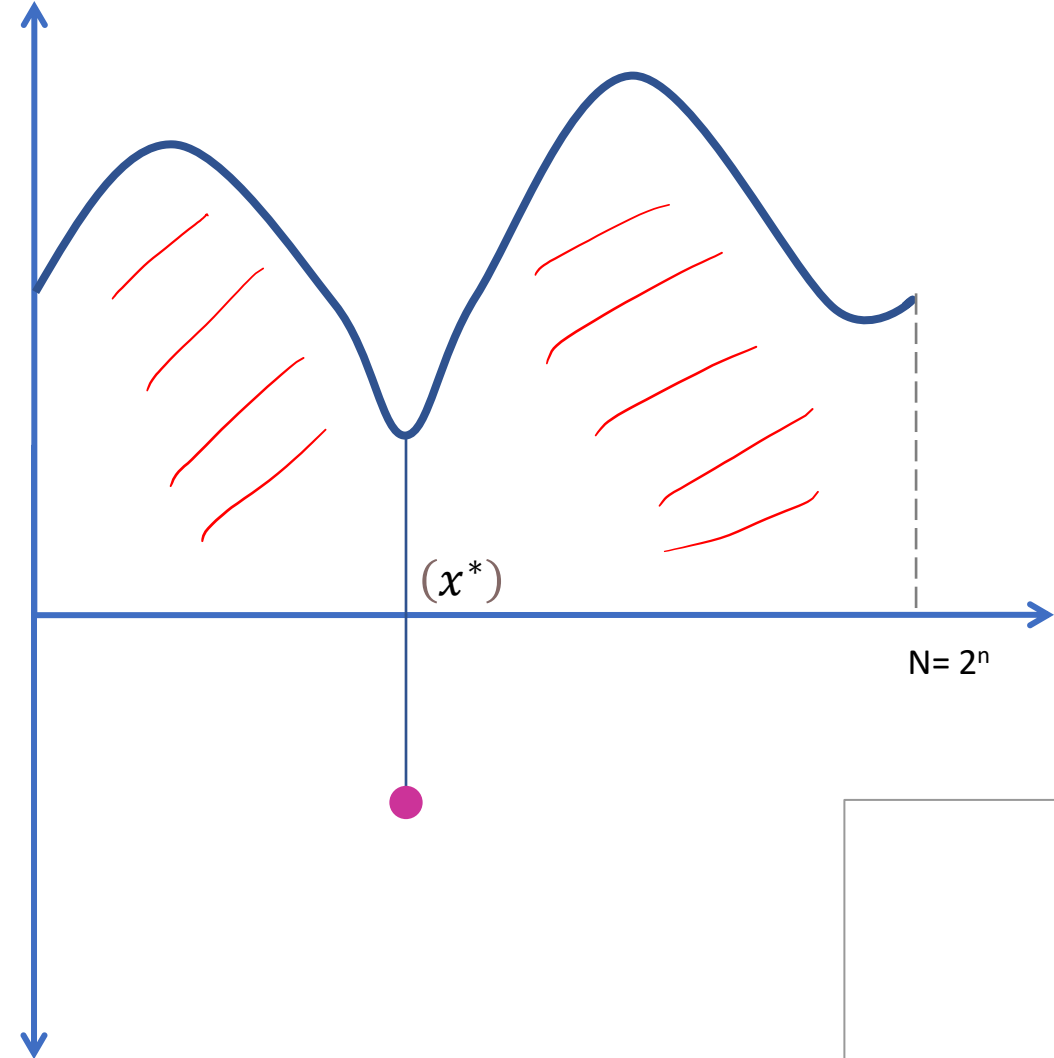
Two steps: 1. Phase inversion

$$\sum_x \alpha_x |x\rangle$$



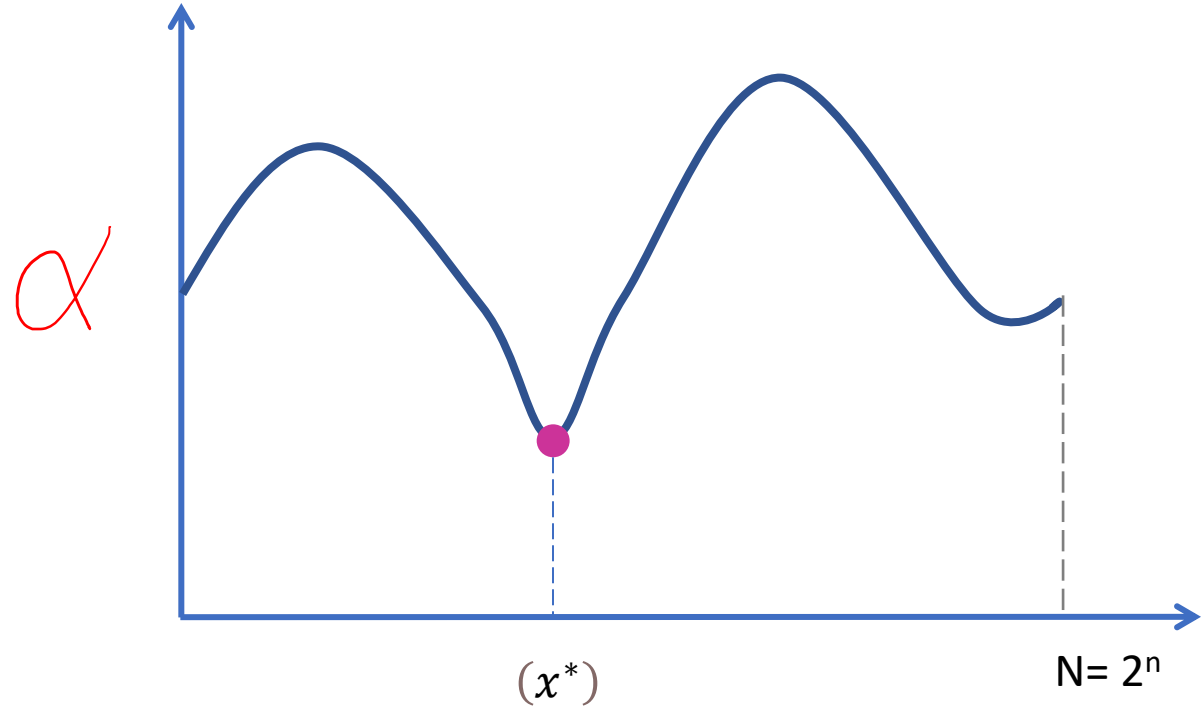
$$\sum_{x \neq x^*} \alpha_x |x\rangle - \alpha_{x^*} |x^*\rangle$$

Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)



Two steps: 2. Inversion over mean

$$\sum_x \alpha_x |x\rangle$$



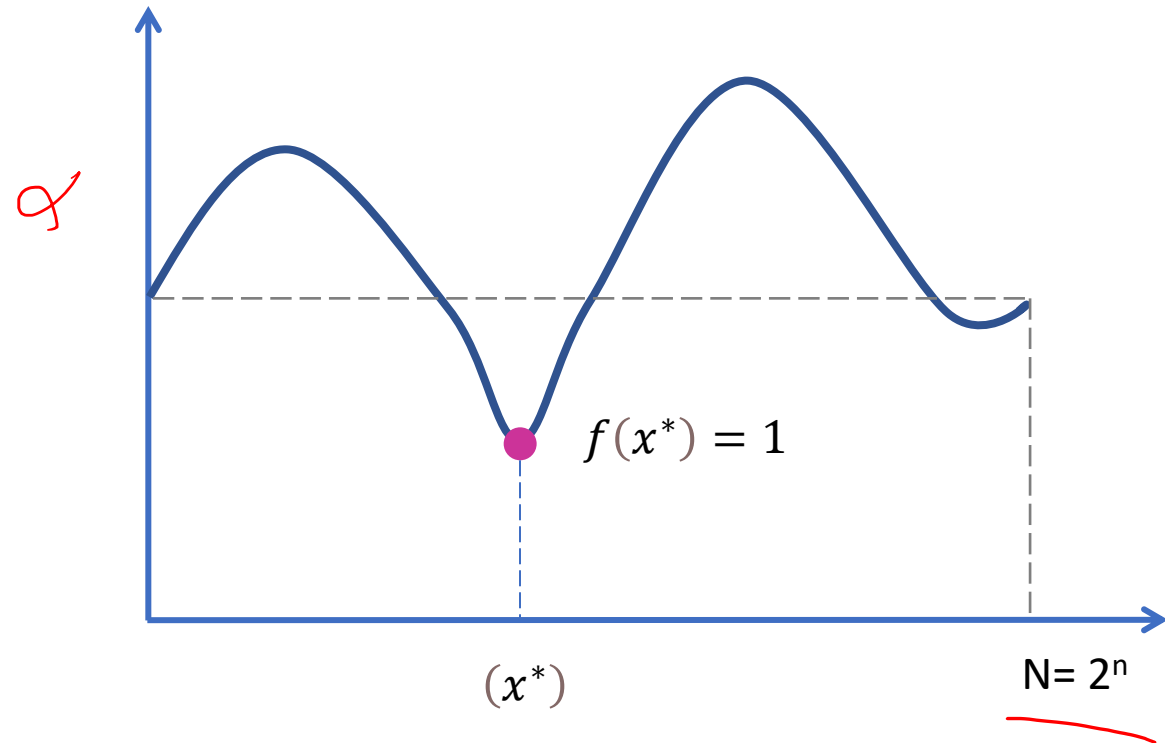
$$f(x^*) = 1$$

Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

Two steps: 2. Inversion over mean

Mean of the amplitudes:

$$\mu = \frac{\sum_{x=0}^{N-1} \alpha_x}{N}$$



Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

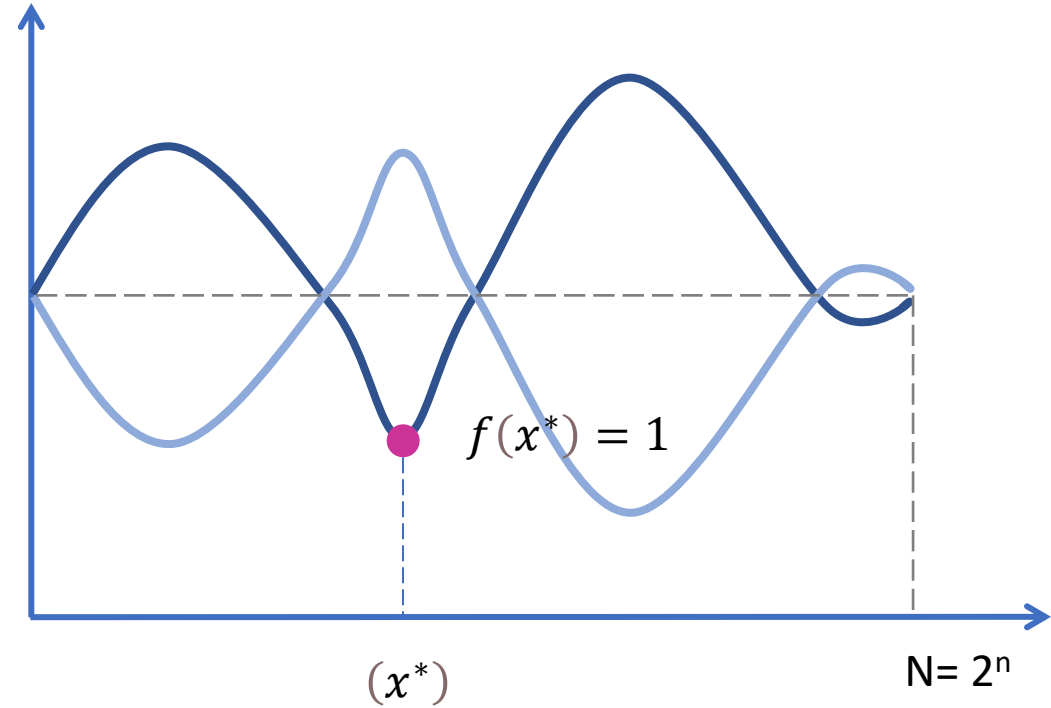
Two steps: 2. Inversion over mean

Mean of the amplitudes:

$$\mu = \frac{\sum_{x=0}^{N-1} \alpha_x}{N}$$

$$\alpha_x \rightarrow 2\mu - \alpha_x$$

$$\sum_x \alpha_x |x\rangle \rightarrow \sum_x (2\mu - \alpha_x) |x\rangle$$



Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

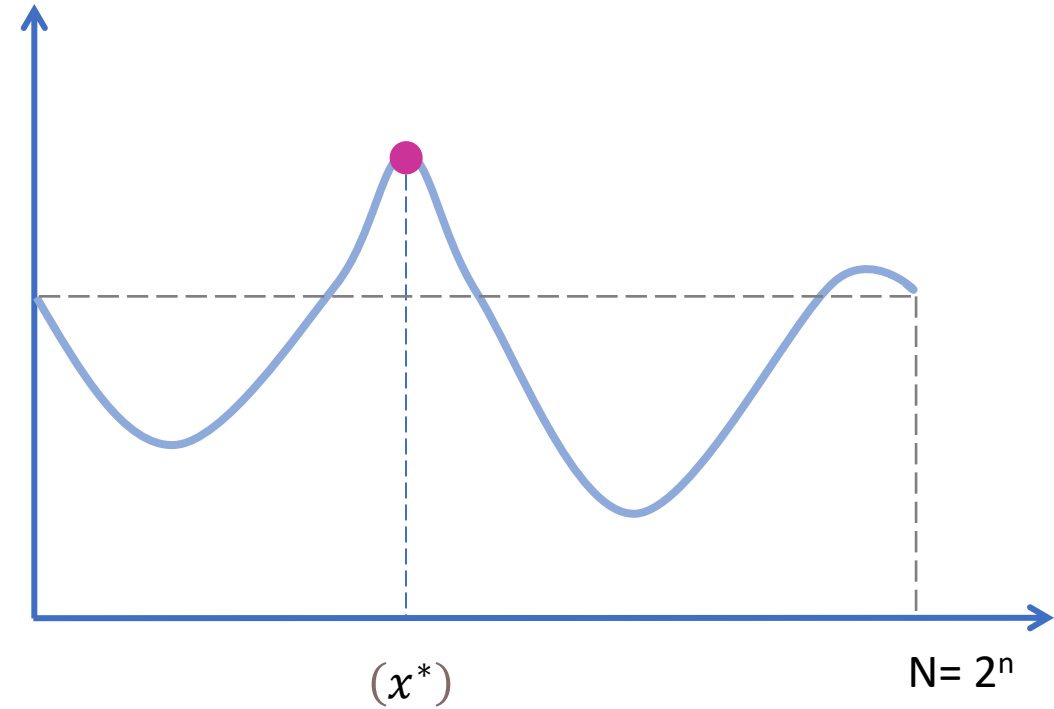
Two steps: 2. Inversion over mean

Mean of the amplitudes:

$$\mu = \frac{\sum_{x=0}^{N-1} \alpha_x}{N}$$

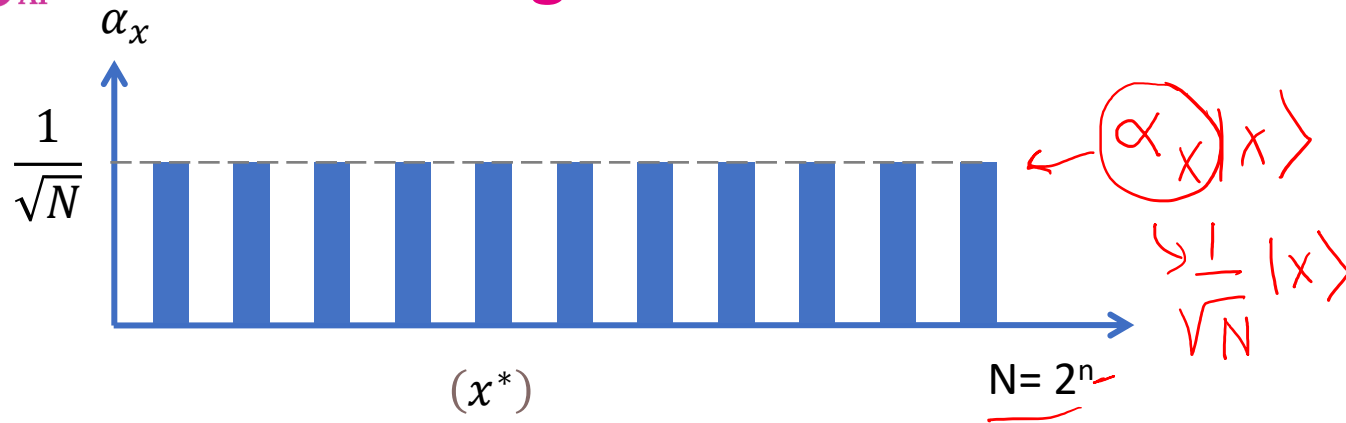
$$\alpha_x \rightarrow 2\mu - \alpha_x$$

$$\sum_x \alpha_x |x\rangle \longrightarrow \sum_x (2\mu - \alpha_x) |x\rangle$$



Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

Grover's algo:



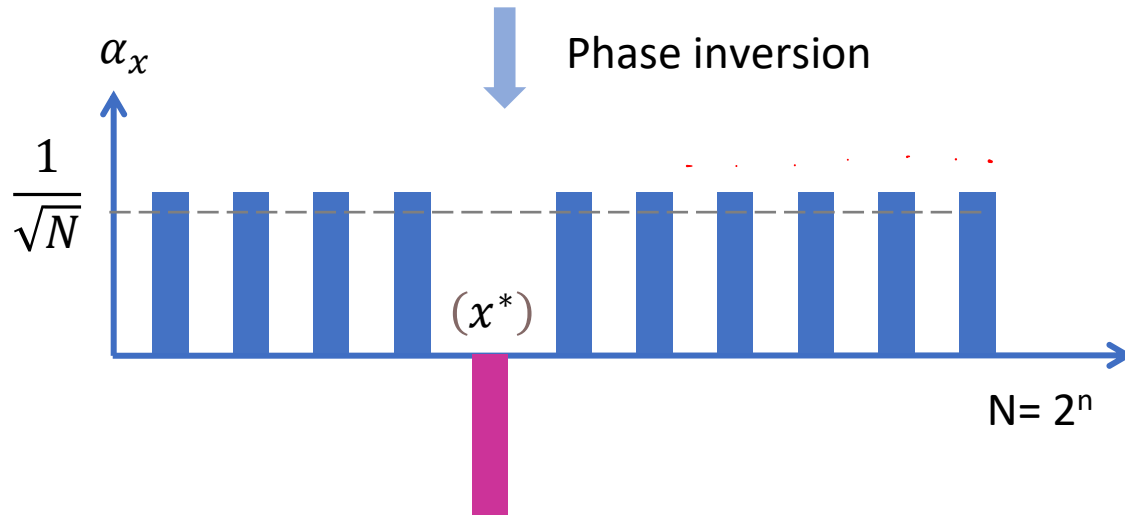
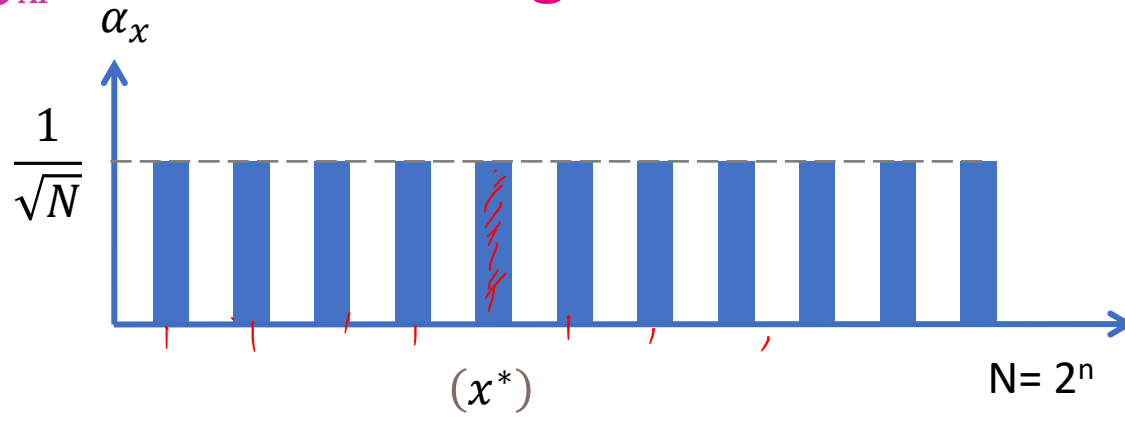
Problem Statement:

$$f: \{0, 1, 2, \dots, N-2, N-1\} \rightarrow \{0, 1\}$$

Find $x: \underline{f(x) = 1}$

Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

Grover's algo:



Problem Statement:

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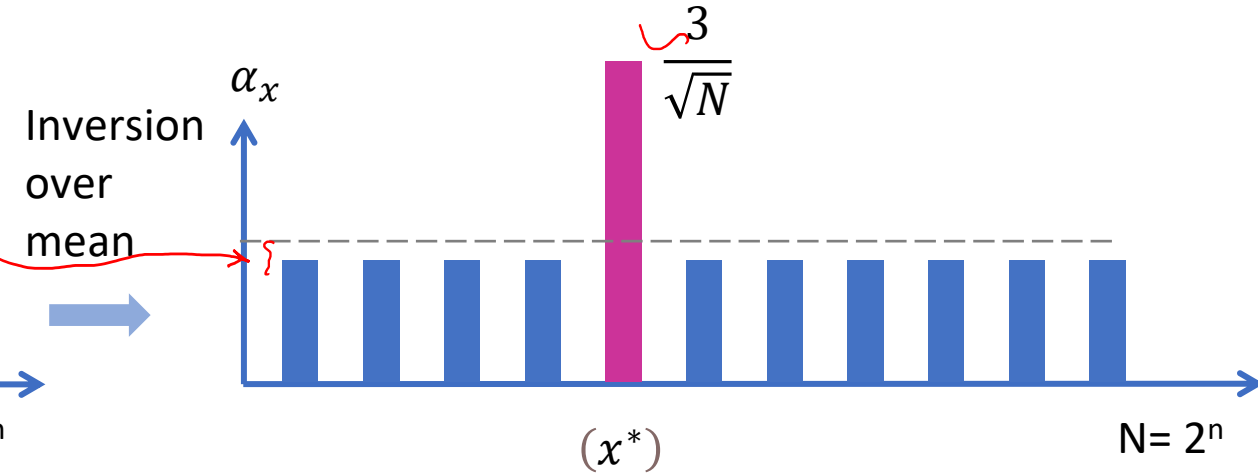
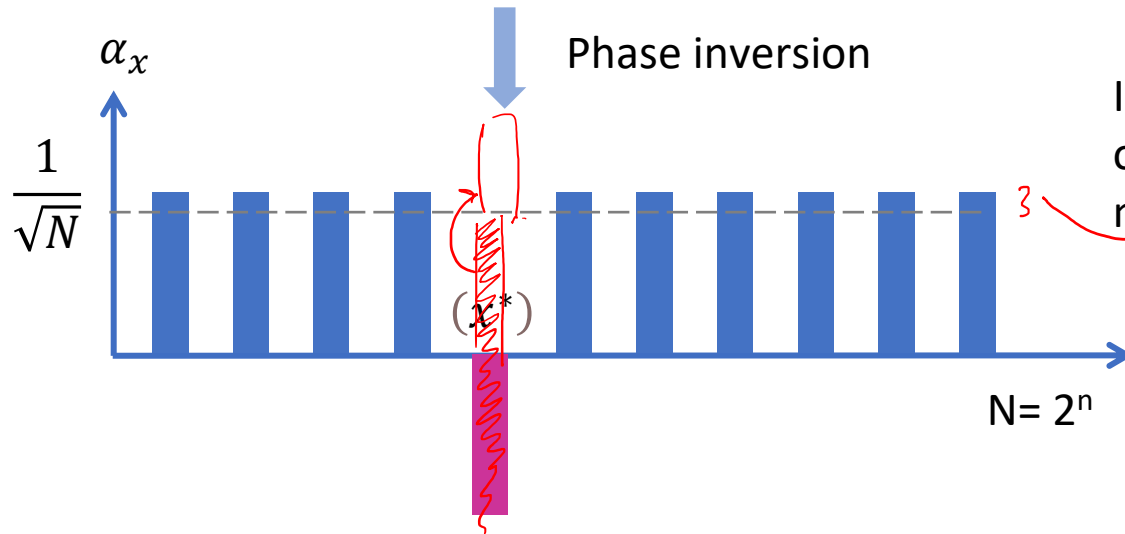
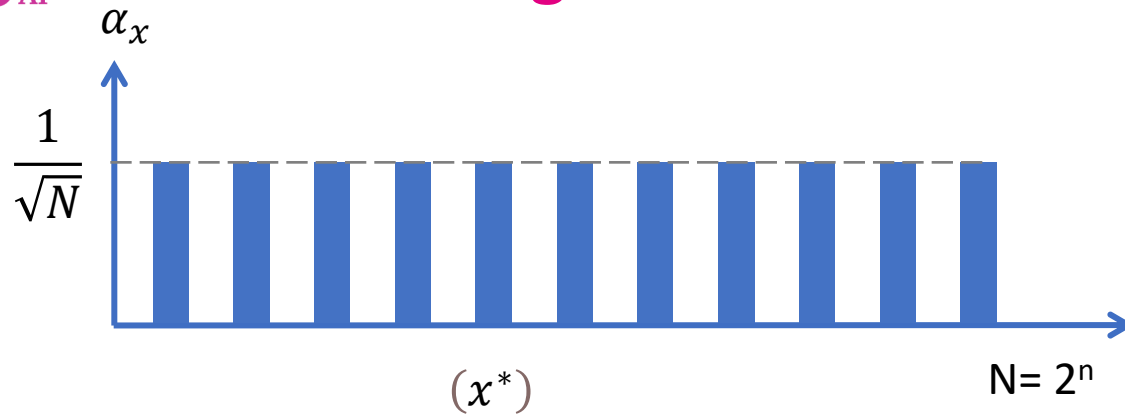
Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

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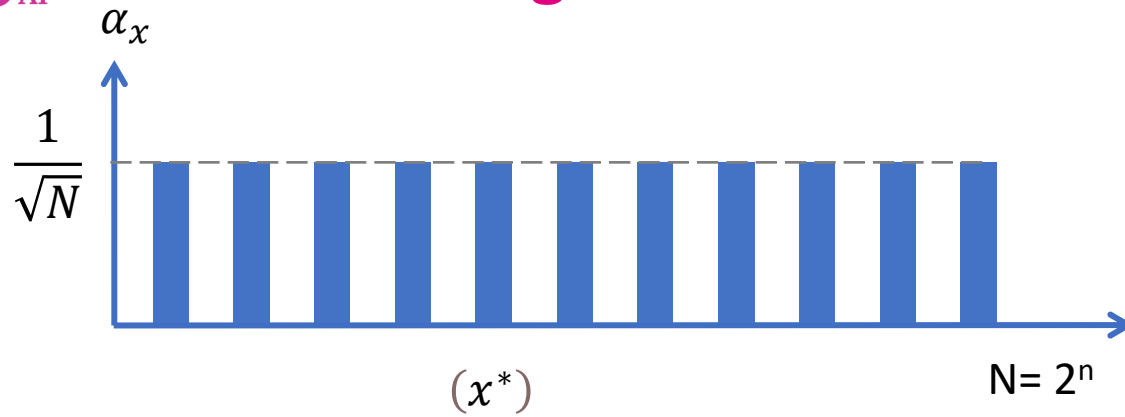
Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

Grover's algo:

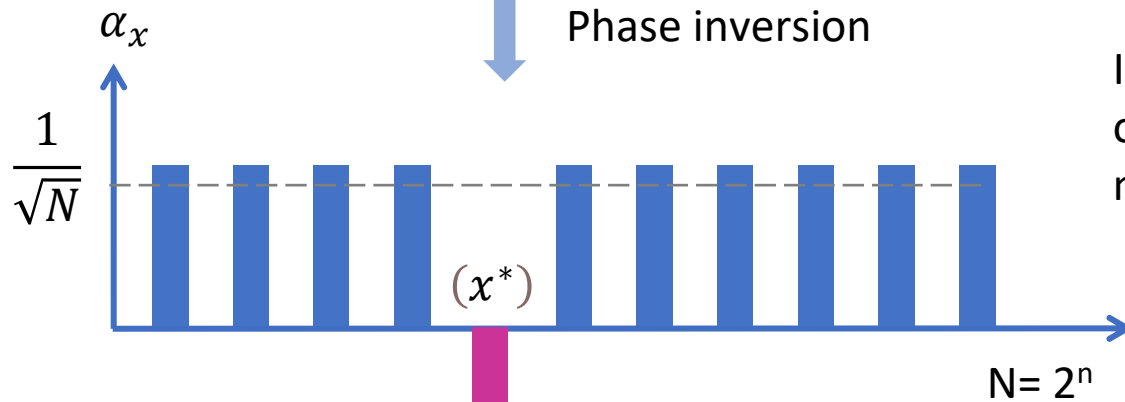
Problem Statement:

$$f: \{0, 1, 2, \dots, N-2, N-1\} \rightarrow \{0, 1\}$$

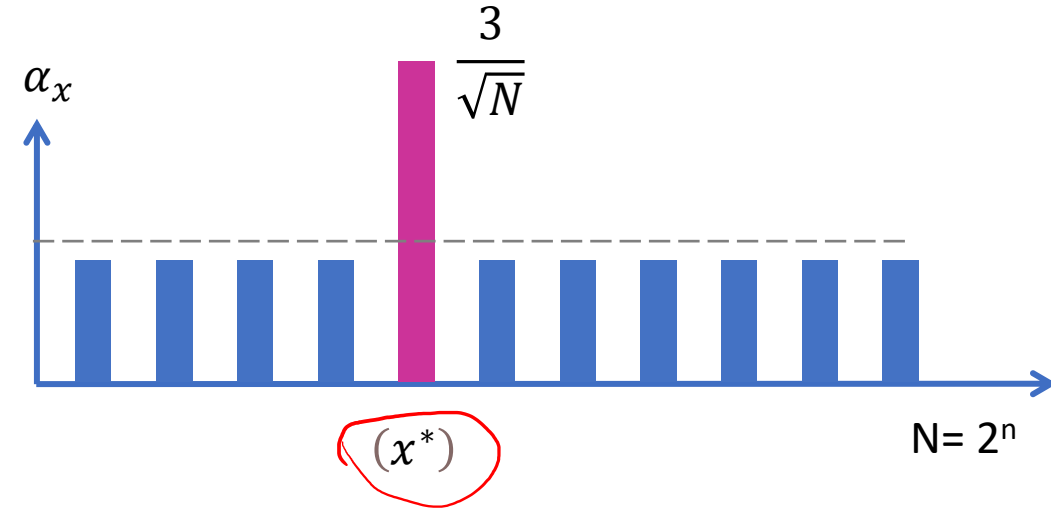
Find $x: f(x) = 1$



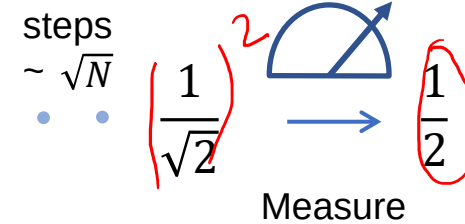
Phase inversion



Inversion over mean

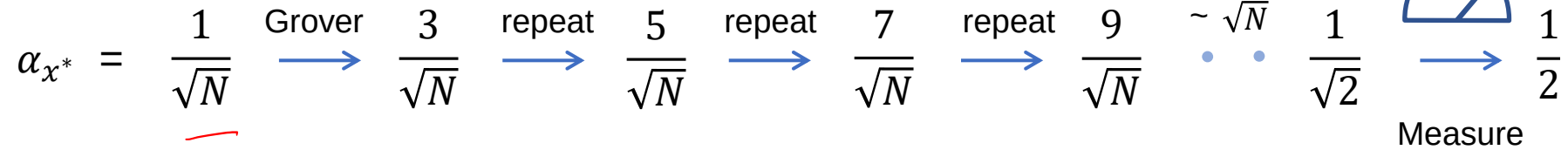


$$\alpha_{x^*} = \frac{1}{\sqrt{N}} \xrightarrow{\text{Grover}} \frac{3}{\sqrt{N}} \xrightarrow{\text{repeat}} \frac{5}{\sqrt{N}} \xrightarrow{\text{repeat}} \frac{7}{\sqrt{N}} \xrightarrow{\text{repeat}} \frac{9}{\sqrt{N}}$$



Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

Grover's algo:

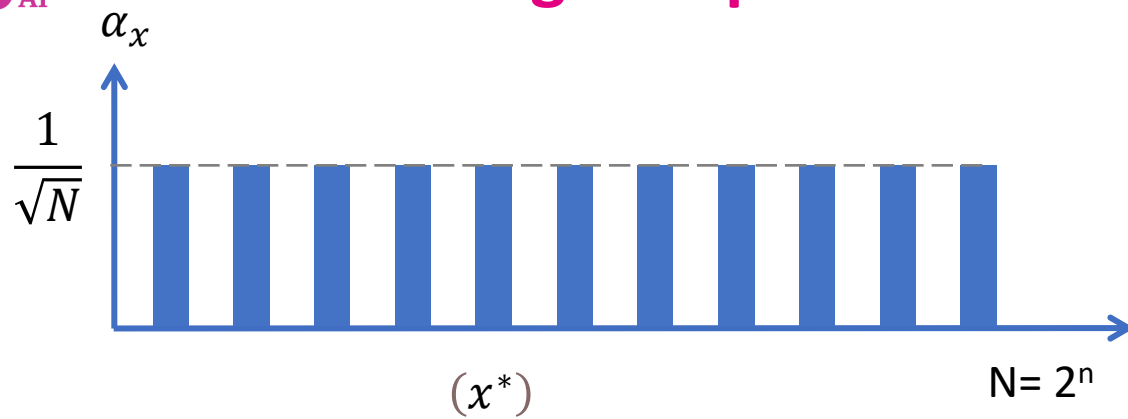


What about others : α_x $\sim \frac{1}{\sqrt{2N}}$ After $\sqrt{N}$ steps

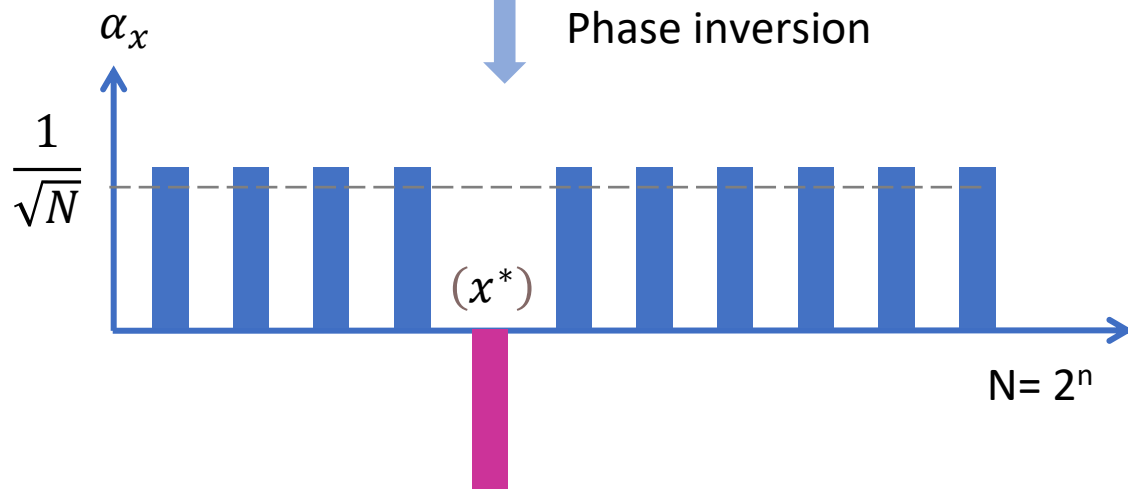


$$f(x^*) = 1$$

Q π_{AI} Grover's algo: Implementation



Phase inversion

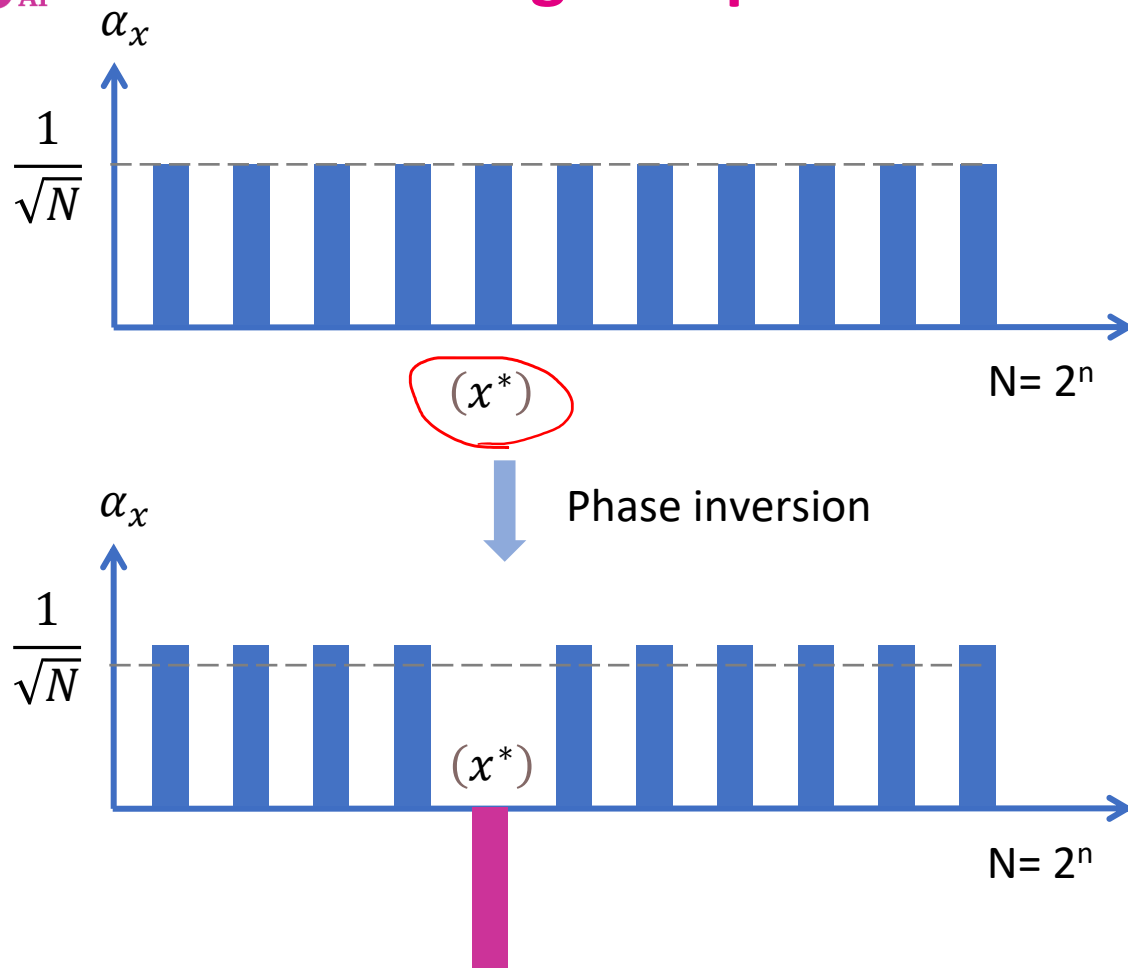


$$\sum_x \alpha_x |x\rangle$$

$$\sum_{x \neq x^*} \alpha_x |x\rangle - \alpha_{x^*} |x^*\rangle$$

Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

Q π_{AI} Grover's algo: Implementation



$$\sum_x \alpha_x |x\rangle$$

Handwritten notes in red:

$$f(x^*) = 1$$

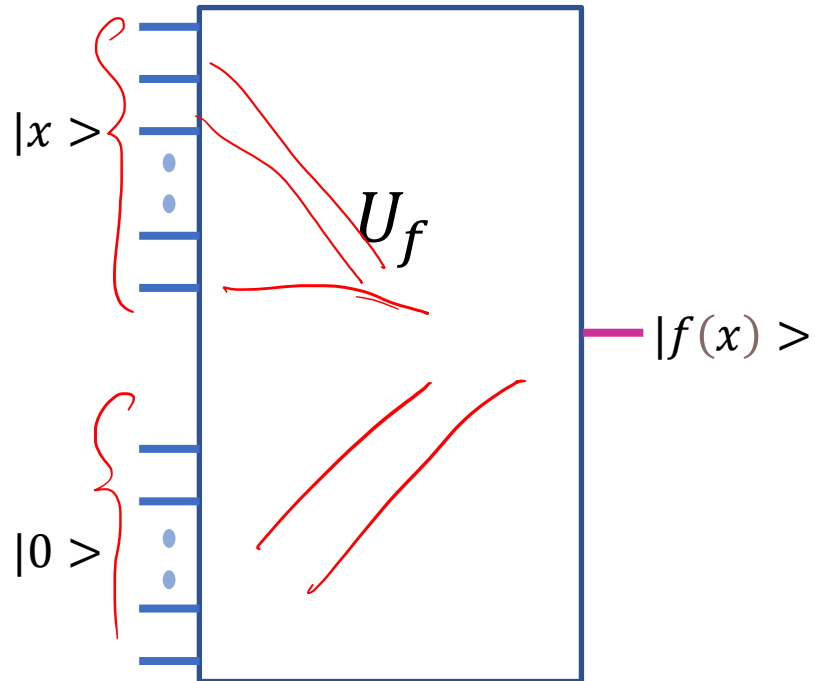
$$(-1)^{f(x)} = \begin{cases} -1 & x = x^* \\ 1 & x \neq x^* \end{cases}$$

$$\sum_x \alpha_x (-1)^{f(x)} |x\rangle$$

Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

Grover's algo: the 'Oracle'

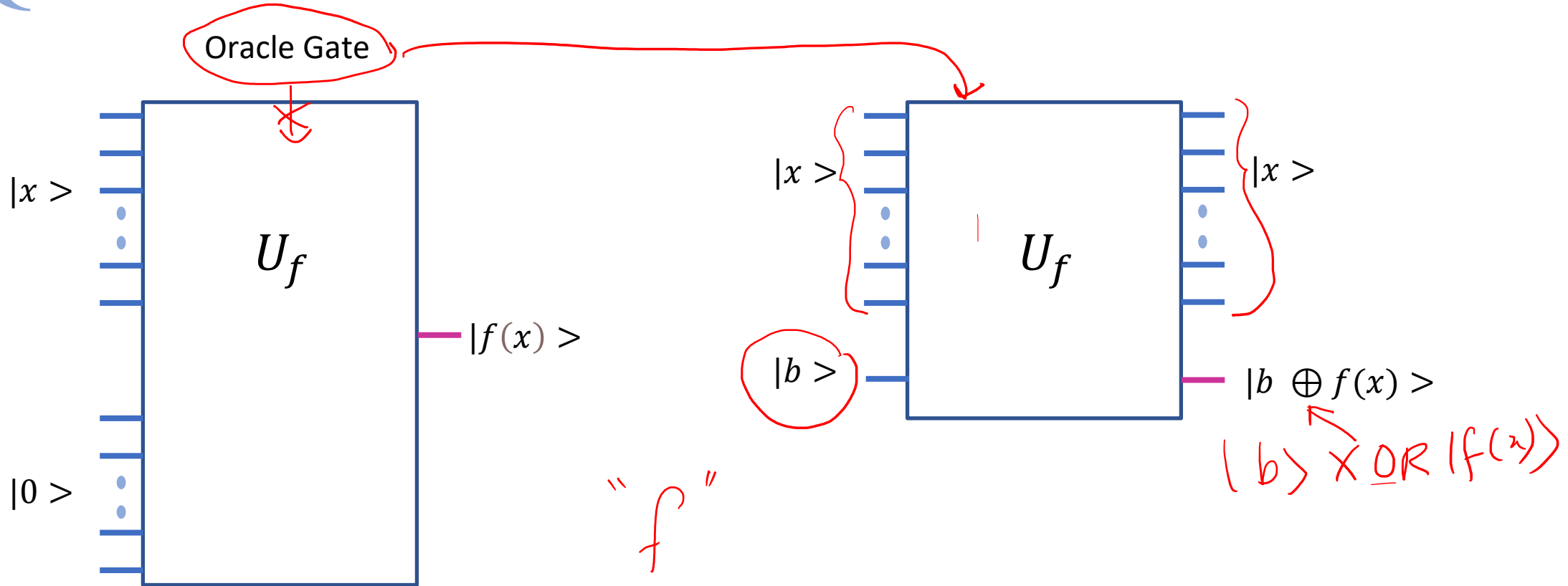
Oracle Gate



Not valid,
Non reversible
Should be unitary

Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

Grover's algo: the 'Oracle'

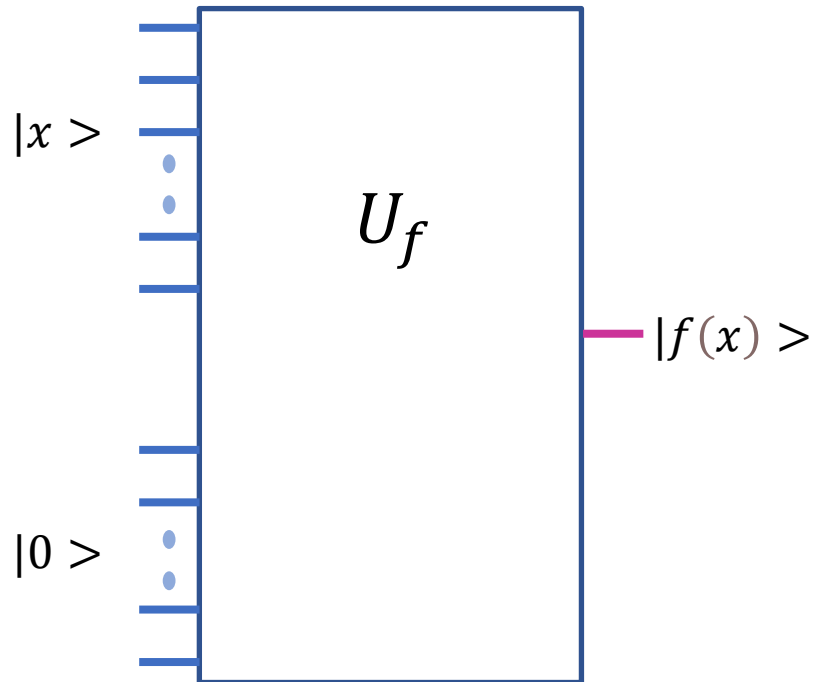


Not valid,
Non reversible
Should be unitary

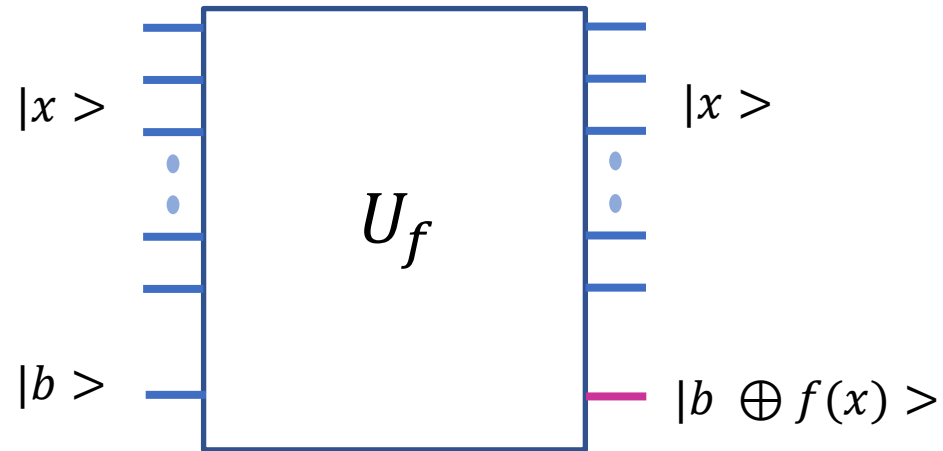
Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

Grover's algo: the 'Oracle'

Oracle Gate



Not valid,
Non reversible
Should be unitary



$|f(x)\rangle$
 $x \neq x^*$ x^*

	0	1
$ b\rangle$ 0	0	1
1	1	0
	$ -\rangle$	$- -\rangle$

$$\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$\rightarrow |-\rangle$$

Handwritten notes:

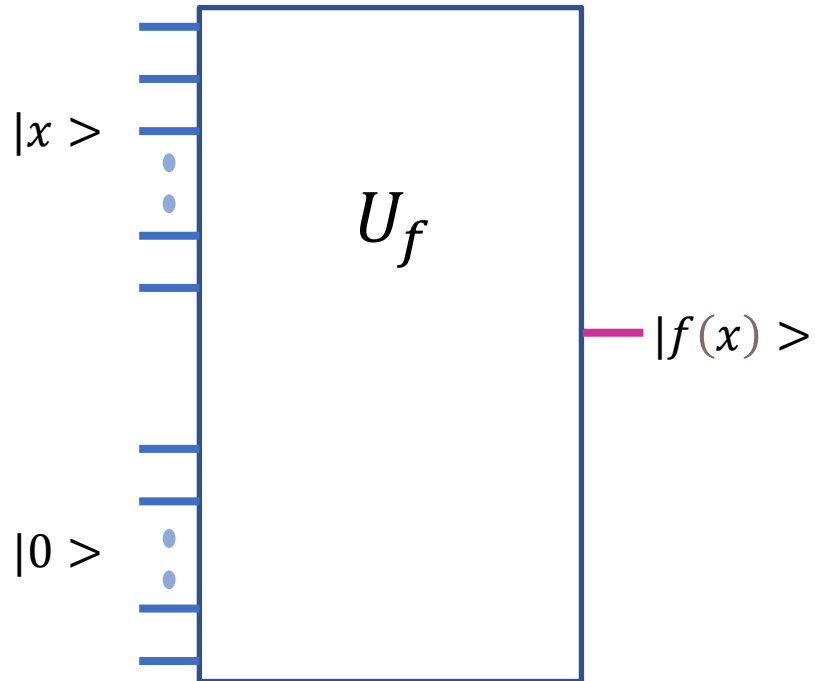
$$f(x \neq x^*) \rightarrow \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$f(x^*) \rightarrow \frac{1}{\sqrt{2}}(|1\rangle - |0\rangle) = -\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

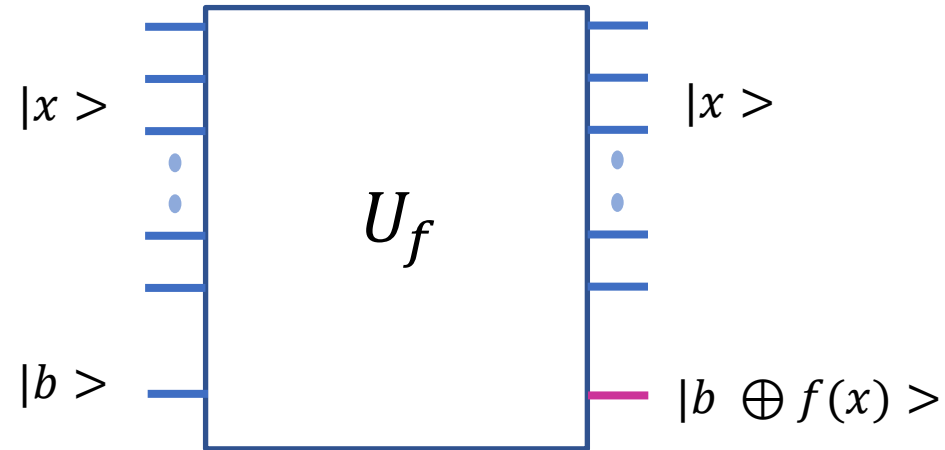
Grover's algo: the 'Oracle'

Oracle Gate



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$|f(x)\rangle$

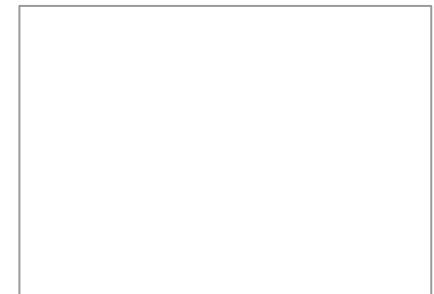
$|b\rangle$

	0	1
0	0	1
1	1	0
$ -\rangle$	$ -\rangle$	$- -\rangle$

$$\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

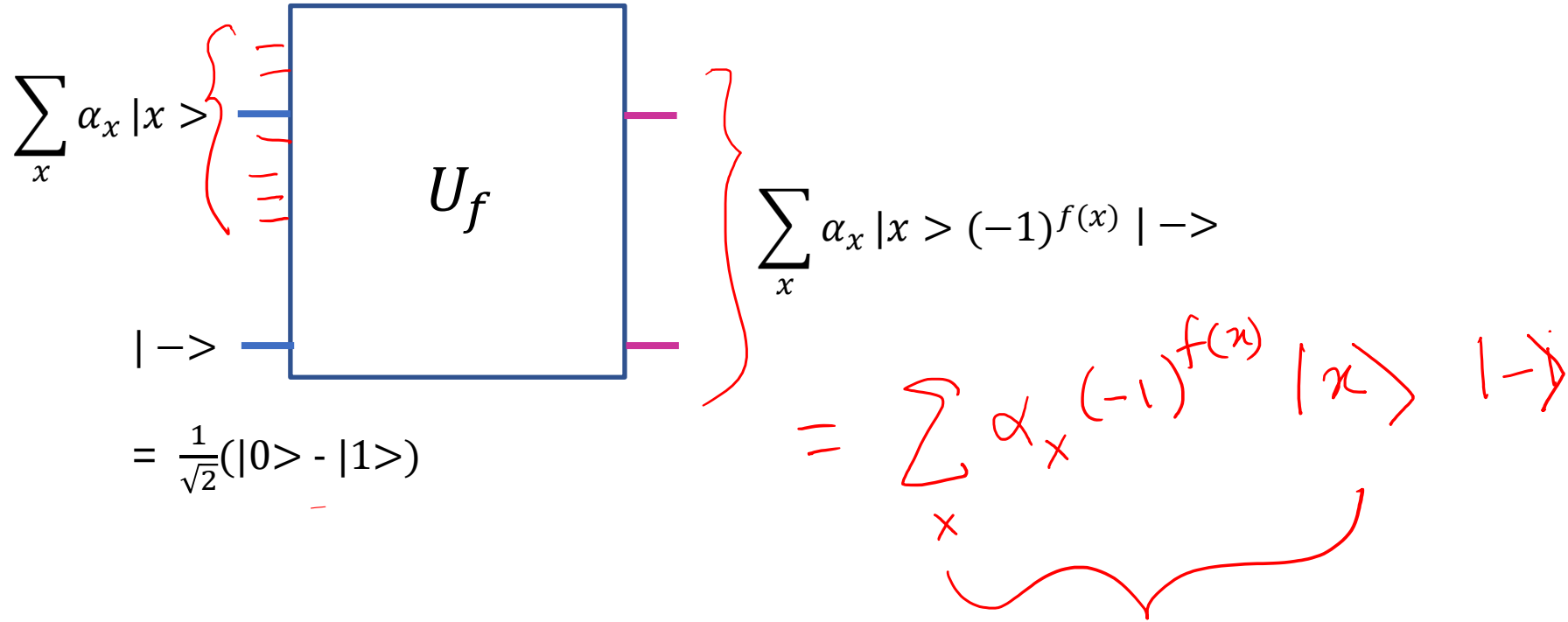
$|-\rangle$

$$= (-1)^{f(x)} |-\rangle$$



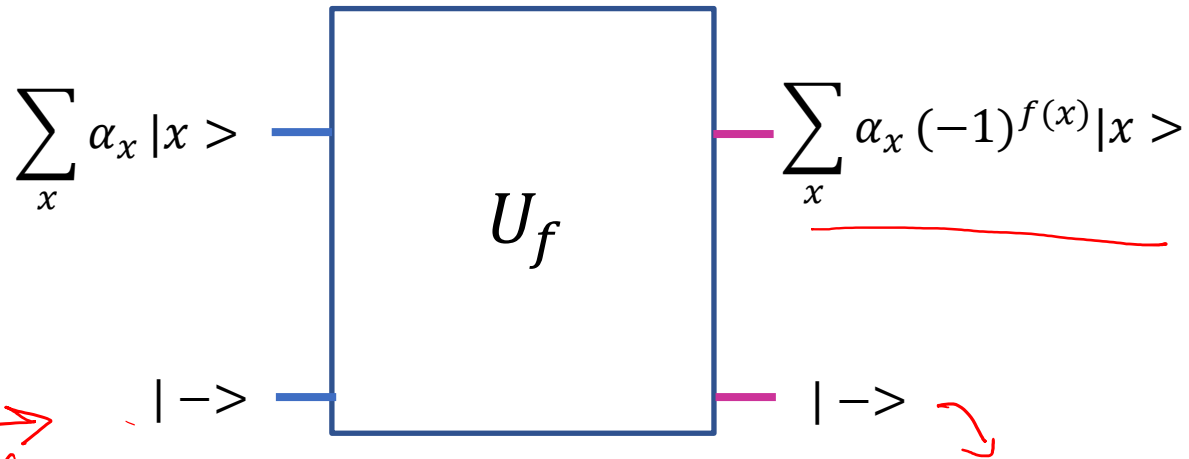
Grover's algo: the 'Oracle'

Phase Oracle



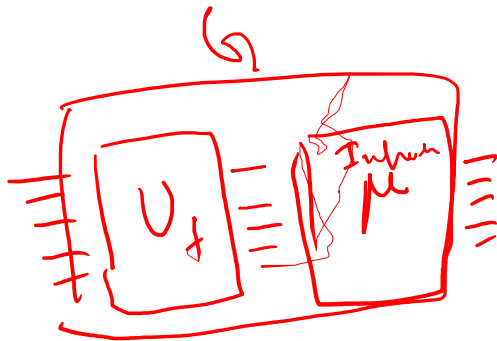
Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

Grover's algo: the 'Oracle'



$\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$
 U_2

Phase inversion !

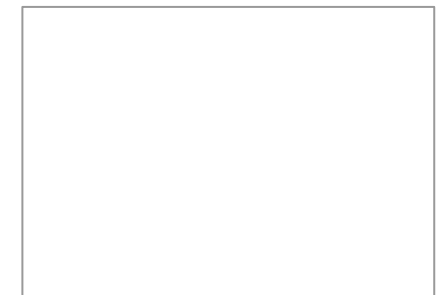


Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

$$\sum_x \alpha_x |x\rangle$$



$$\sum_x \alpha_x (-1)^{f(x)} |x\rangle$$



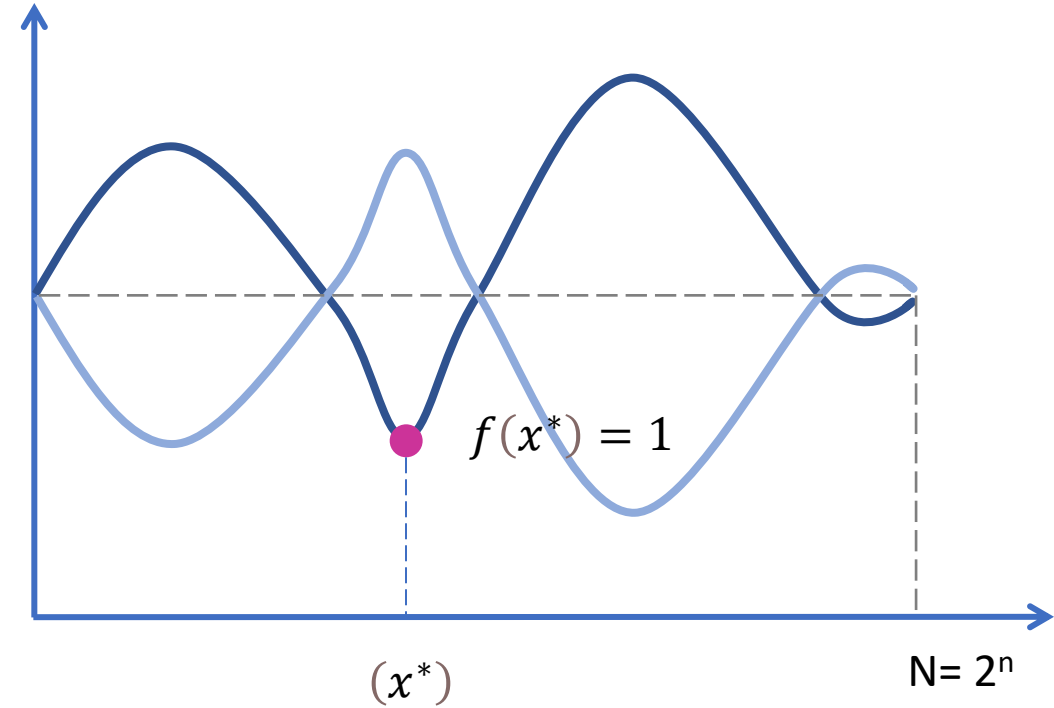
Step 2. Inversion over mean

Mean of the amplitudes:

$$\mu = \frac{\sum_{x=0}^{N-1} \alpha_x}{N}$$

$$\alpha_x \rightarrow 2\mu - \alpha_x$$

$$\sum_x \alpha_x |x\rangle \longrightarrow \sum_x (2\mu - \alpha_x) |x\rangle$$

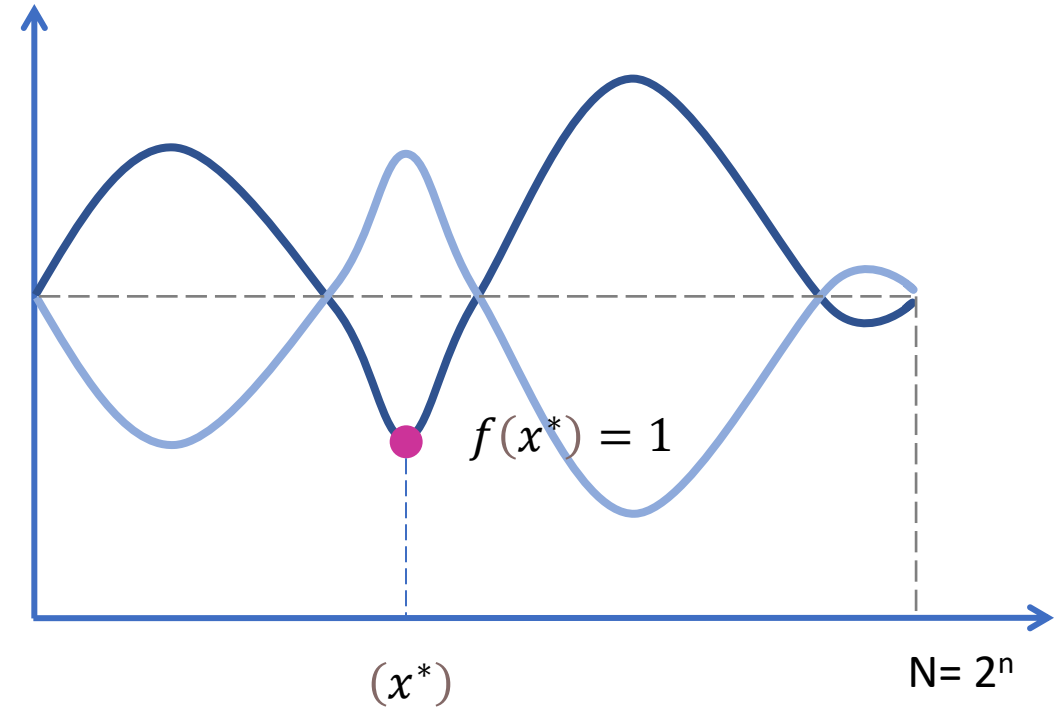


Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

Step 2. Inversion over mean

Inversion over mean =
Reflection about the state:

$$|u\rangle = \frac{1}{\sqrt{N}} \sum_x |x\rangle$$

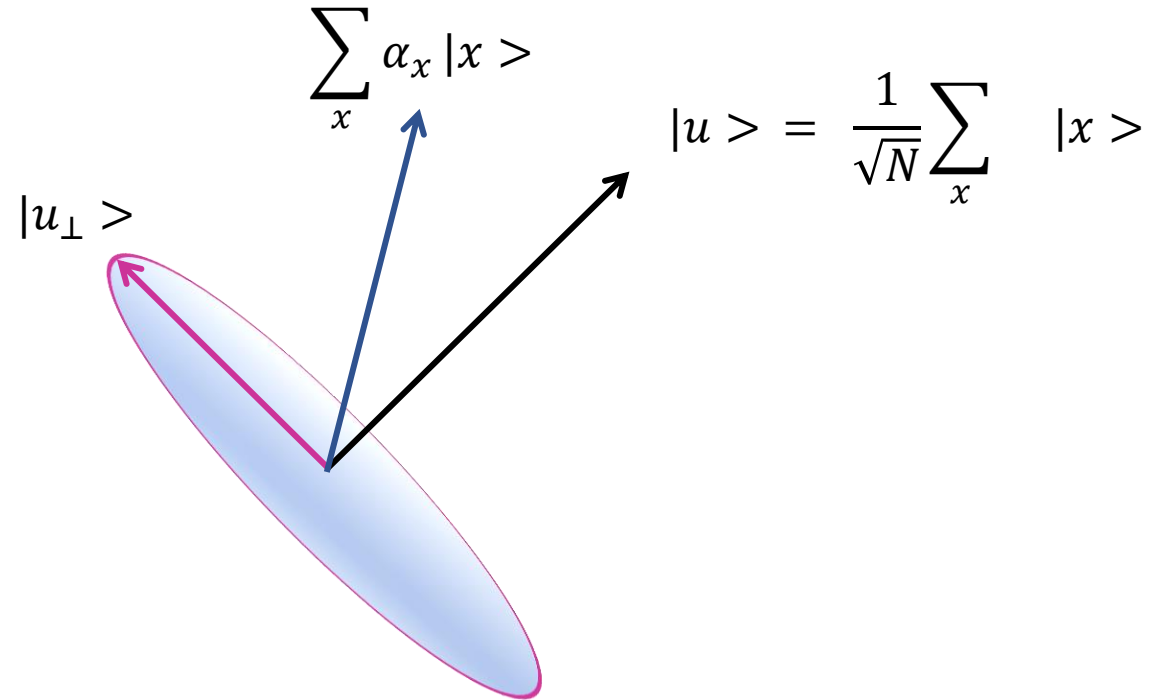


Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

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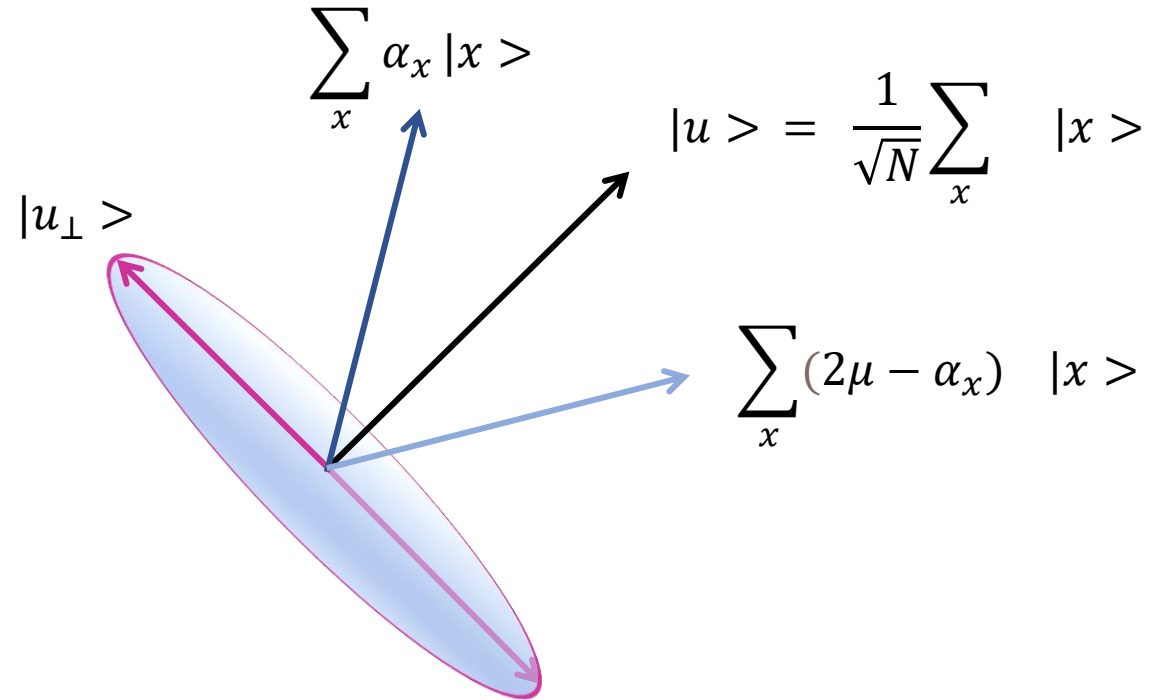


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Step 2. Inversion over mean

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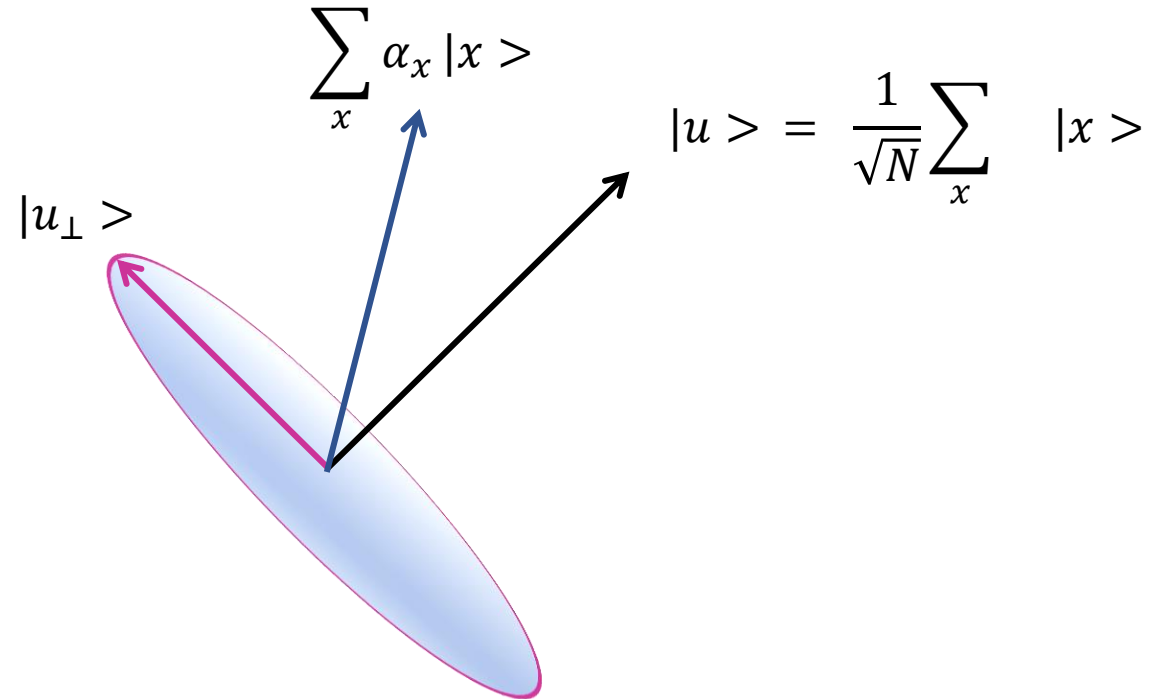
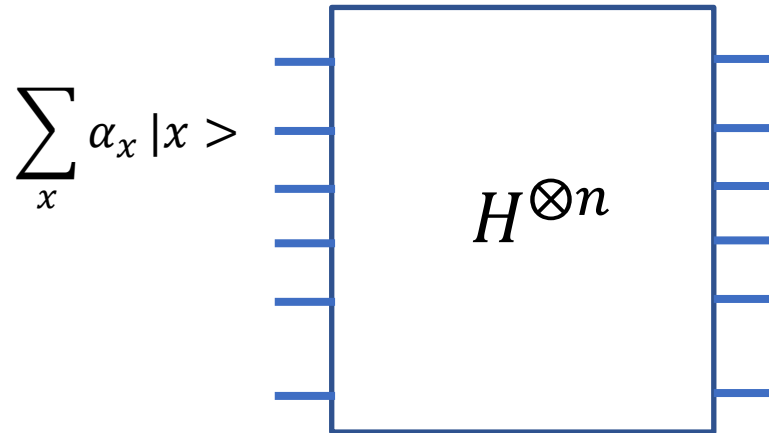
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Step 2. Inversion over mean

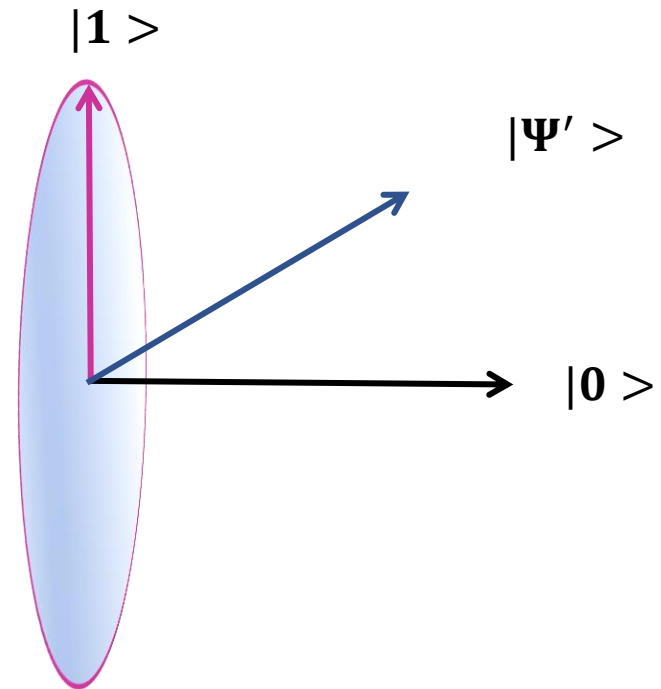
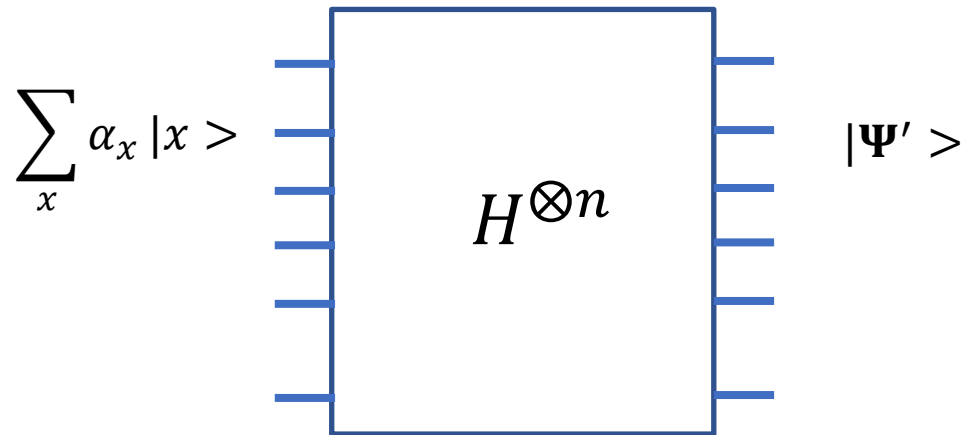
Hadamard over n qubits



Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

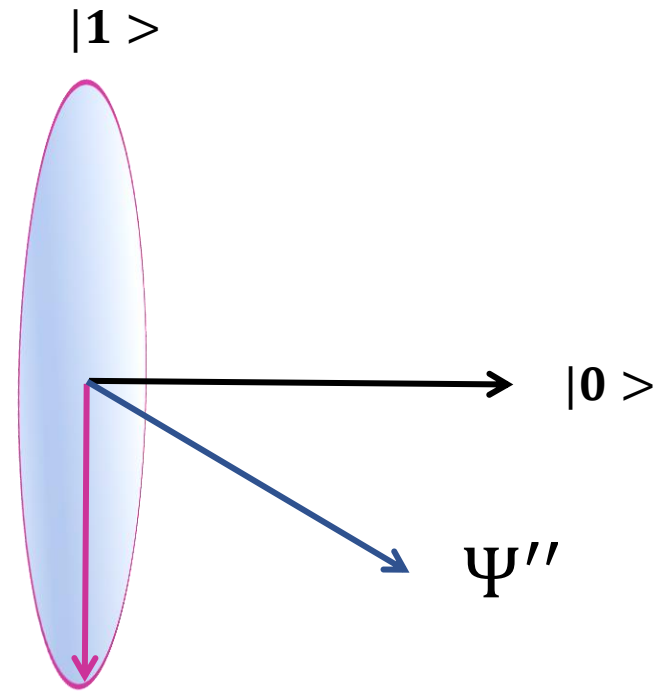
Step 2. Inversion over mean

Hadamard over n qubits



Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

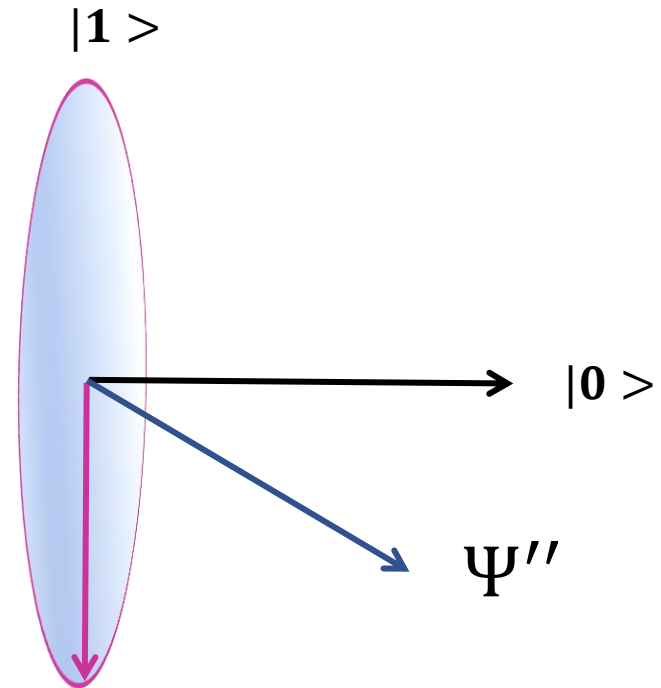
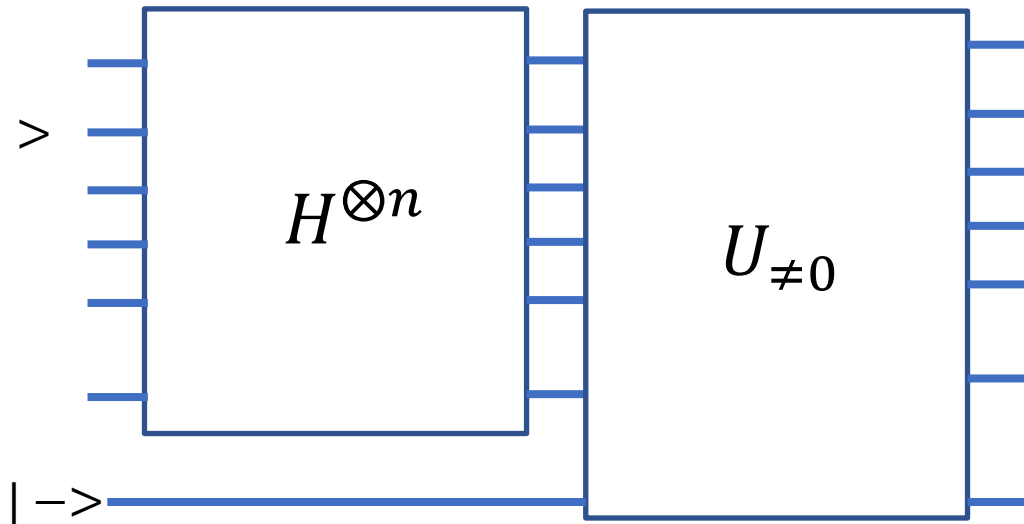
Step 2. Inversion over mean



Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

Step 2. Inversion over mean

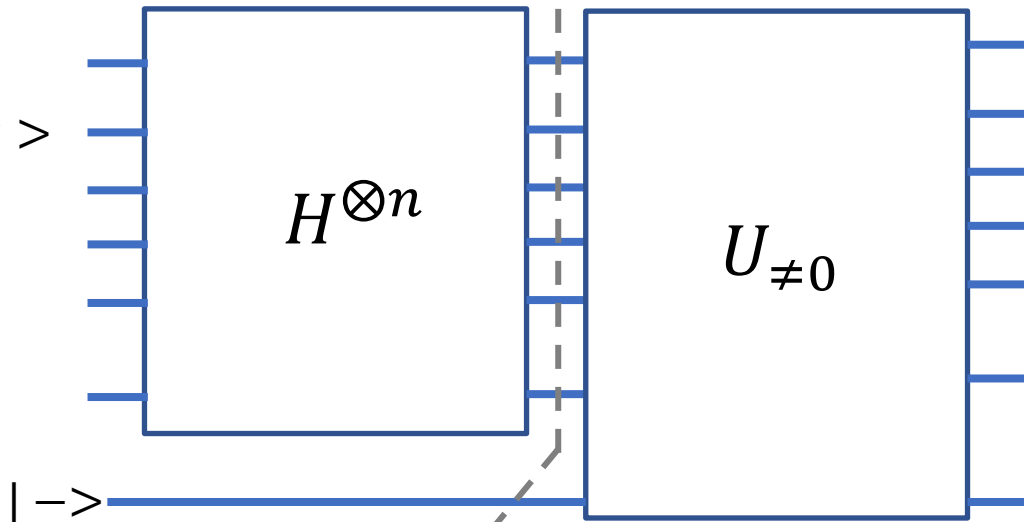
Hadamard over n qubits



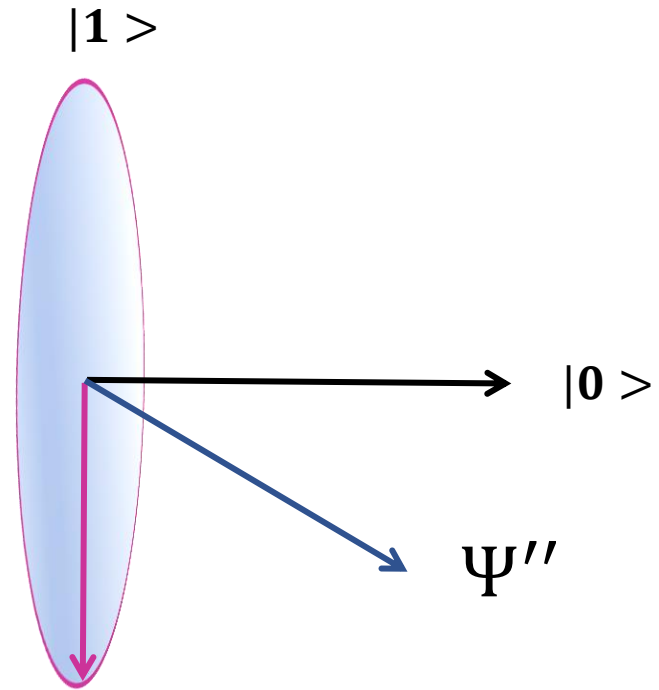
Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

Step 2. Inversion over mean

Hadamard over n qubits Reflection about $|0\rangle$



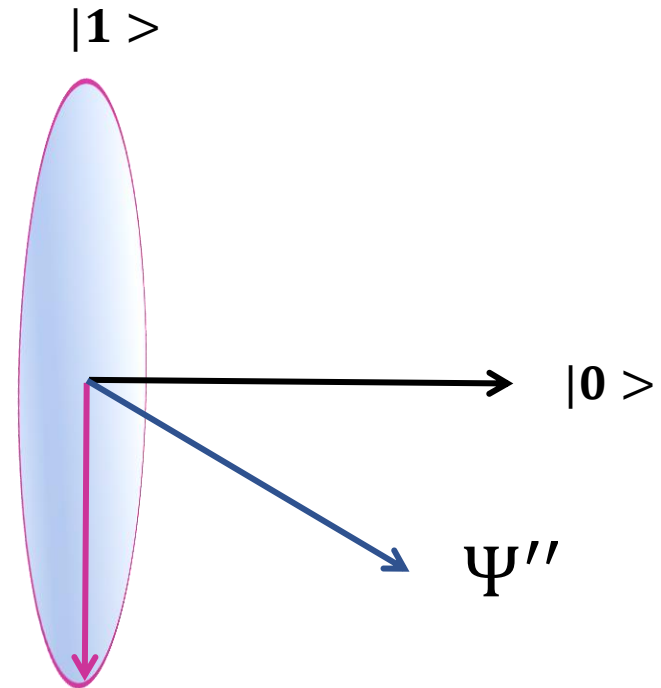
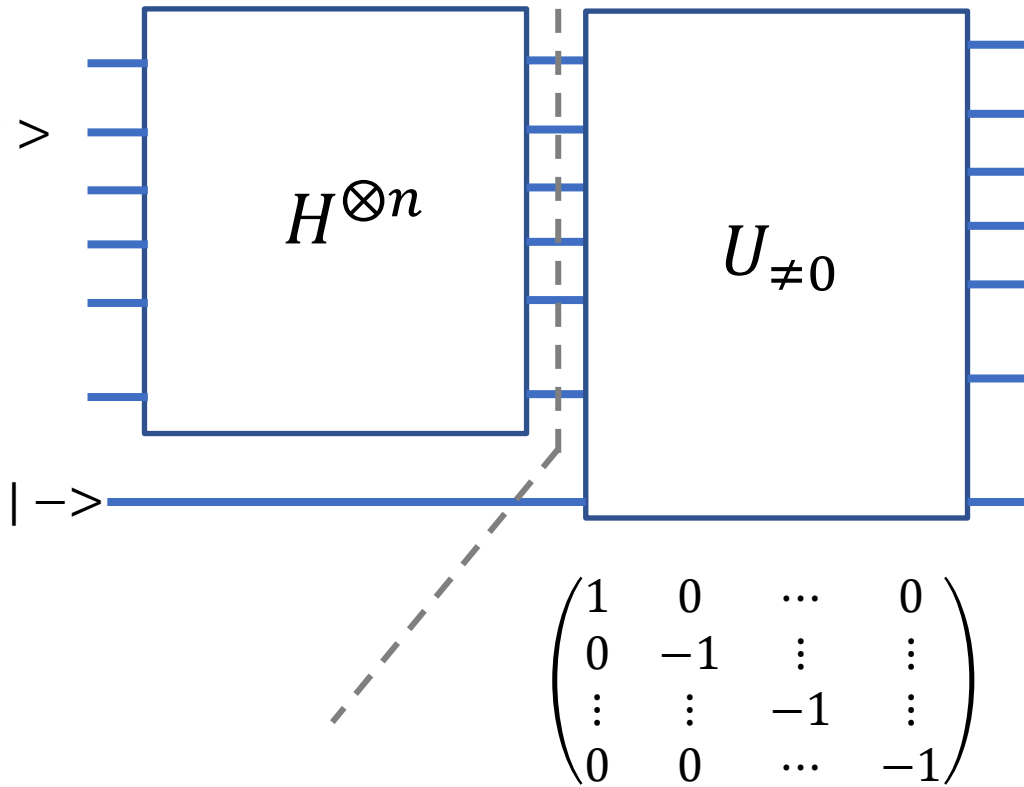
If the input is All 0 then =0
Else multiply -1!



Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

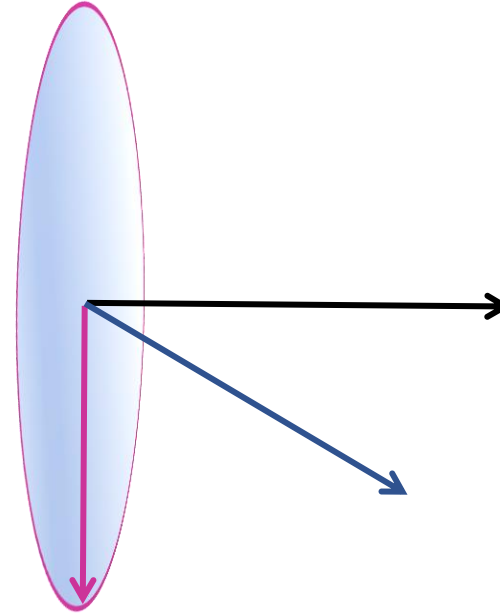
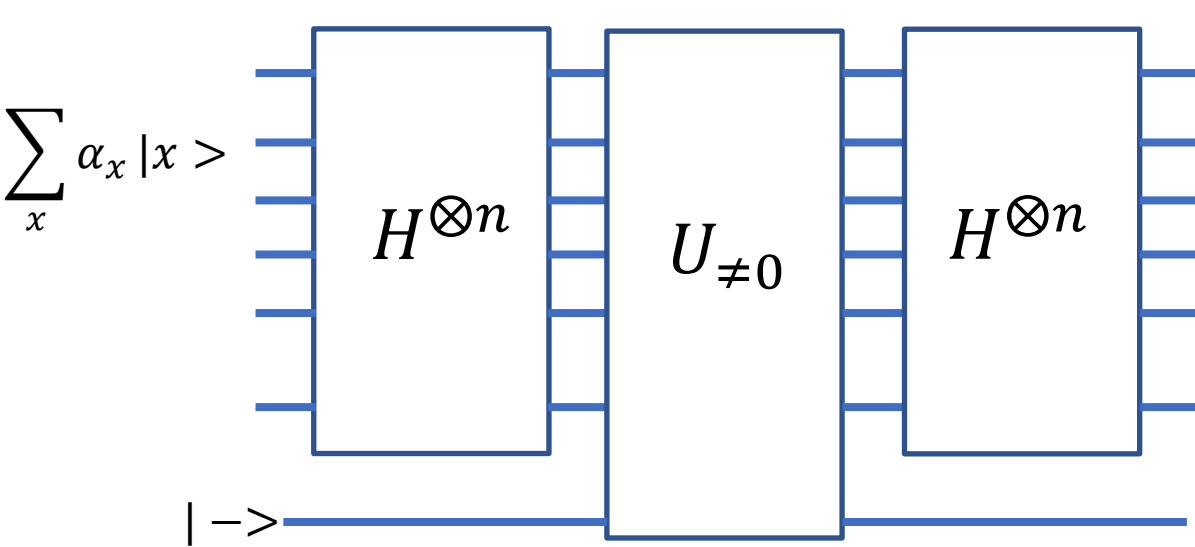
Step 2. Inversion over mean

Hadamard over n qubits Reflection about $|0\rangle$



Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

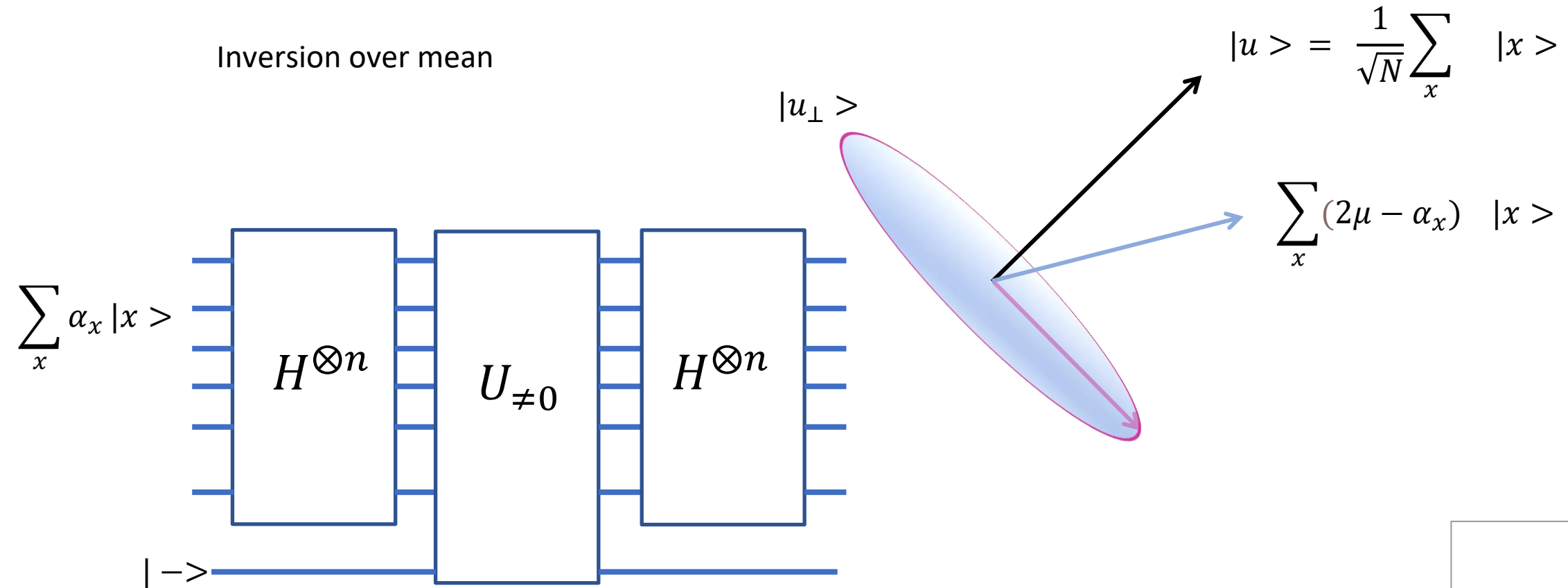
Step 2. Inversion over mean



Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

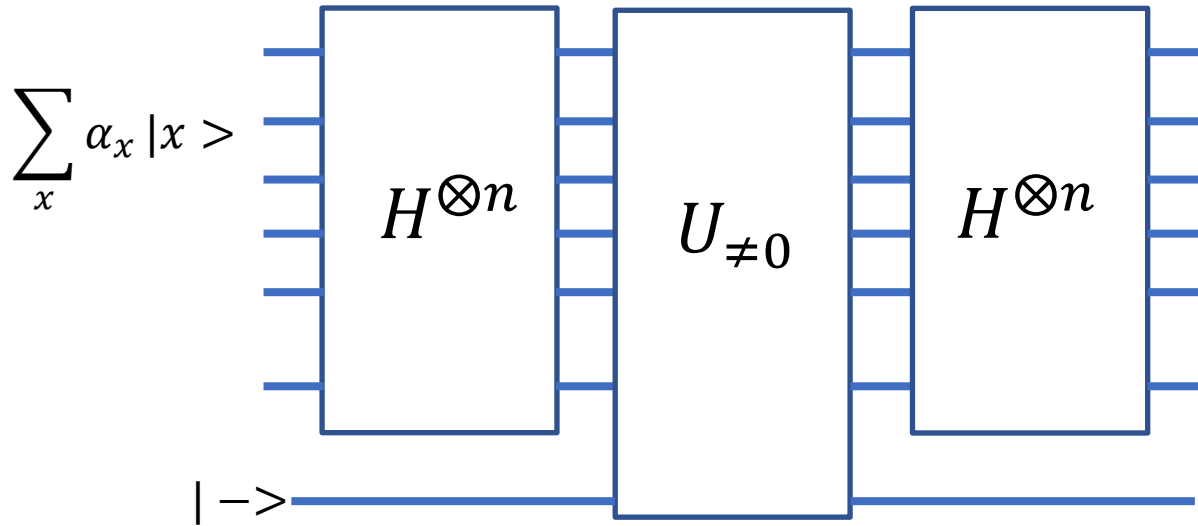
Step 2. Inversion over mean

Inversion over mean



Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

Step 2. Inversion over mean



$$H^{\otimes n} \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & -1 & \vdots & \vdots \\ \vdots & \vdots & -1 & \vdots \\ 0 & 0 & \cdots & -1 \end{pmatrix} H^{\otimes n}$$

Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

Step 2. Inversion over mean

$$H^{\otimes n} \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & -1 & \vdots & \vdots \\ \vdots & \vdots & -1 & \vdots \\ 0 & 0 & \cdots & -1 \end{pmatrix} H^{\otimes n}$$

$$= H^{\otimes n} \left[\begin{pmatrix} 2 & 0 & \cdots & 0 \\ 0 & 0 & \vdots & \vdots \\ \vdots & \vdots & 0 & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix} - \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \vdots & \vdots \\ \vdots & \vdots & 1 & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} \right] H^{\otimes n}$$

$$= H^{\otimes n} \left[\begin{pmatrix} 2 & 0 & \cdots & 0 \\ 0 & 0 & \vdots & \vdots \\ \vdots & \vdots & 0 & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix} - I \right] H^{\otimes n}$$

Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

Step 2. Inversion over mean

$$\begin{aligned}
 &= H^{\otimes n} \left[\begin{pmatrix} 2 & 0 & \dots & 0 \\ 0 & 0 & \vdots & \vdots \\ \vdots & \vdots & 0 & \vdots \\ 0 & 0 & \dots & 0 \end{pmatrix} - I \right] H^{\otimes n} \\
 &= H^{\otimes n} \begin{pmatrix} 2 & 0 & \dots & 0 \\ 0 & 0 & \vdots & \vdots \\ \vdots & \vdots & 0 & \vdots \\ 0 & 0 & \dots & 0 \end{pmatrix} H^{\otimes n} - H^{\otimes n} I H^{\otimes n} \\
 &= H^{\otimes n} \begin{pmatrix} 2 & 0 & \dots & 0 \\ 0 & 0 & \vdots & \vdots \\ \vdots & \vdots & 0 & \vdots \\ 0 & 0 & \dots & 0 \end{pmatrix} H^{\otimes n} - I = \begin{pmatrix} \frac{2}{\sqrt{N}} & 0 & \dots & 0 \\ \frac{2}{\sqrt{N}} & 0 & \vdots & \vdots \\ \frac{2}{\sqrt{N}} & \vdots & 0 & \vdots \\ \frac{2}{\sqrt{N}} & 0 & \dots & 0 \end{pmatrix} H^{\otimes n} - I
 \end{aligned}$$

Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

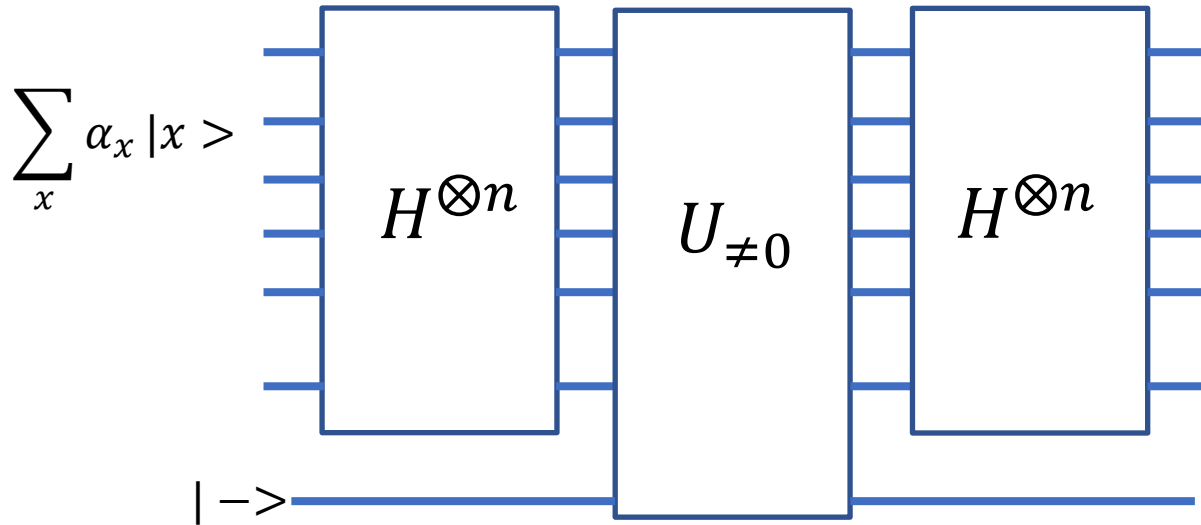
Step 2. Inversion over mean

$$= \begin{pmatrix} \frac{2}{\sqrt{N}} & 0 & \cdots & 0 \\ \frac{2}{\sqrt{N}} & 0 & \vdots & \vdots \\ \vdots & \vdots & 0 & \vdots \\ \frac{2}{\sqrt{N}} & 0 & \cdots & 0 \end{pmatrix} H^{\otimes n} - I = \begin{pmatrix} \frac{2}{N} & \frac{2}{N} & \cdots & \frac{2}{N} \\ \frac{2}{N} & \frac{2}{N} & \vdots & \vdots \\ \vdots & \vdots & \frac{2}{N} & \vdots \\ \frac{2}{N} & \frac{2}{N} & \cdots & \frac{2}{N} \end{pmatrix} - I$$

$$= \begin{pmatrix} \frac{2}{N} - 1 & \frac{2}{N} & \cdots & \frac{2}{N} \\ \frac{2}{N} & \frac{2}{N} - 1 & \vdots & \vdots \\ \vdots & \vdots & \frac{2}{N} - 1 & \vdots \\ \frac{2}{N} & \frac{2}{N} & \cdots & \frac{2}{N} - 1 \end{pmatrix}$$

Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

Step 2. Inversion over mean



$$\begin{pmatrix} \frac{2}{N} - 1 & \frac{2}{N} & \cdots & \frac{2}{N} \\ \frac{2}{N} & \frac{2}{N} - 1 & \vdots & \vdots \\ \vdots & \vdots & \frac{2}{N} - 1 & \vdots \\ \frac{2}{N} & \frac{2}{N} & \cdots & \frac{2}{N} - 1 \end{pmatrix} \begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \vdots \\ \vdots \\ \vdots \\ \alpha_{N-1} \end{pmatrix}$$

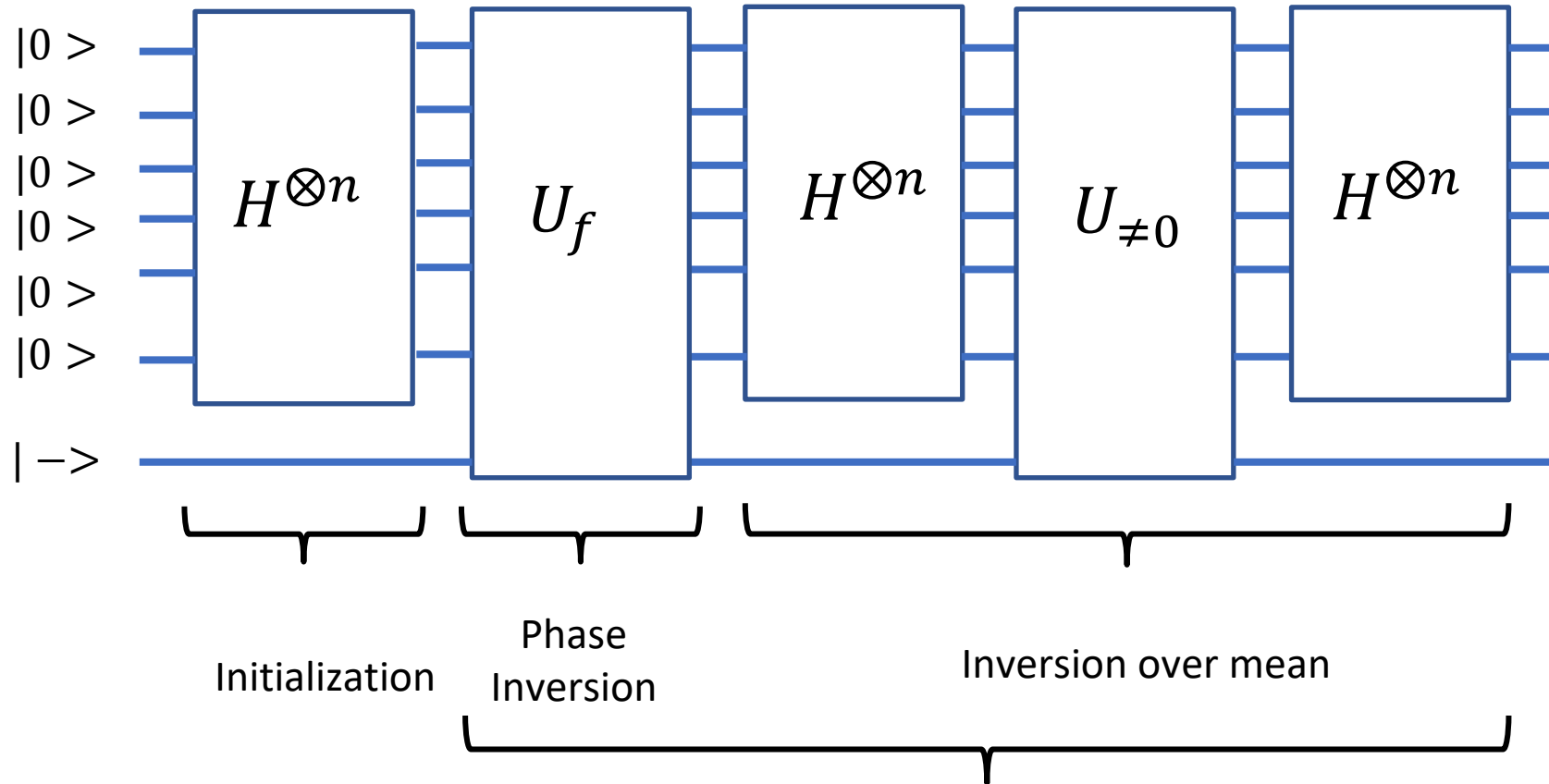
Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

Step 2. Inversion over mean

$$\begin{pmatrix} \frac{2}{N} - 1 & \frac{2}{N} & \cdots & \frac{2}{N} \\ \frac{2}{N} & \frac{2}{N} - 1 & \vdots & \vdots \\ \vdots & \vdots & \frac{2}{N} - 1 & \vdots \\ \frac{2}{N} & \frac{2}{N} & \cdots & \frac{2}{N} - 1 \end{pmatrix} \begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \vdots \\ \alpha_x \\ \vdots \\ \alpha_{N-1} \end{pmatrix} = \begin{pmatrix} \frac{2}{N} \sum_x \alpha_x - \alpha_0 \\ \frac{2}{N} \sum_x \alpha_x - \alpha_1 \\ \vdots \\ \frac{2}{N} \sum_x \alpha_x - \alpha_{x^*} \\ \vdots \\ \frac{2}{N} \sum_x \alpha_x - \alpha_{N-1} \end{pmatrix} \longrightarrow (2\mu - \alpha_x)$$

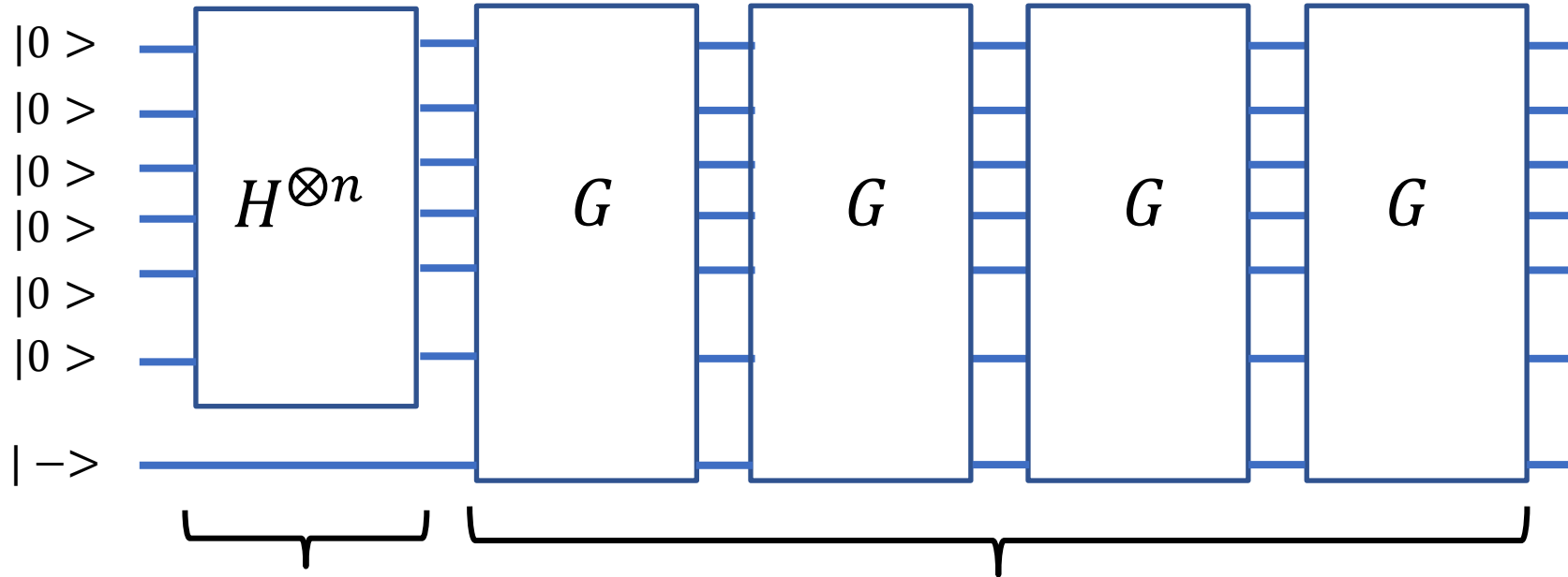
Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

Quantum Circuit: Grover's Algo



Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)

Quantum Circuit: Grover's Algo



$$\alpha_{x^*} = \frac{1}{\sqrt{N}} \xrightarrow{\text{Grover}} \frac{3}{\sqrt{N}} \xrightarrow{\text{repeat}} \frac{5}{\sqrt{N}} \xrightarrow{\text{repeat}} \frac{7}{\sqrt{N}} \xrightarrow{\text{repeat}} \frac{9}{\sqrt{N}} \cdots \frac{1}{\sqrt{2}} \xrightarrow{\text{Measure}} \frac{1}{2}$$

What about others : $\alpha_x \sim \frac{1}{\sqrt{2N}}$ After $\sqrt{N}$ steps

Ref: L K Grover, Phys. Rev. Lett. 78(2), 325 (1997)



Thank you for your attention!

